



Program User Manual

Version 5.0, build 5428.

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Homepage <https://wongzit.github.io/program/pyaroma>

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Preface

py.Aroma 5 is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

If you found any problem or bug, it would be helpful if you can contact me via E-mail (wang.zhe.dr@gmail.com). I am glad to help on your problems and your bug report will help me to make py.Aroma better.

For bug report, please include following information as much as possible:

- (1) OS type and version.
- (2) Source code mode?
- (2) Is the bug/error reproducible? How to reproduce the bug?
- (3) Did the bug make program crashing or give wrong/unnatural result?
- (4) If possible, please attach the file(s) when the bug occurred.

I am also very happy to accept request for new functions and add them to new version of py.Aroma. Please feel free to contact me via E-mail.

If py.Aroma 5 is utilized in your work, or the code is implied in your own code, please consider citing following contents:

- Zhe WANG, py.Aroma: : An Intuitive Graphical User Interface for Diverse Aromaticity Analyses, *Chemistry*, **2024**, 6, (6), 1692-1703. DOI: [10.3390/chemistry6060103](https://doi.org/10.3390/chemistry6060103)



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1 Overview

1.1 About

py.Aroma 5 is a multifunctional tool for aromaticity analysis. It supports bond length alternation (BLA), harmonic oscillator model of aromaticity (HOMA), HOMER (HOMA model for excited states), π orbital axis vector (POAV) and nucleus-independent chemical shift (NICS) analysis. The package also integrates py.NMR¹ and CSigen², enabling NMR spectrum plotting and generation of computational supporting information. In addition, version 5 adds Natural Resonance Theory (NRT) analysis and geometry operation modules.

• Source code layout

pyaroma_main.py – Main code of py.Aroma 5, application entry point.
CONSTANT.py – Constants such as atomic radii and HOMA/HOMER parameters.
readFile.py – Input parsing and geometry extraction.
geomAnalyzer.py – Connectivity detection and ring identification.
NICSInp.py – NICS input file generation.
NICSout.py – Shielding tensor extraction from NICS output.
homaCalc.py – HOMA and HOMER calculations.
pathCreator.py – Path generation for NICS-XY-scan.
poav.py – POAV1 and POAV2 calculations.
pynmr.py – Chemical shift calculation and NMR spectrum plotting.
config.ini – Program settings.
assets – Required images and other static resources.

1.2 Running Program

py.Aroma 5 is written with Python 3.13 and utilizes PyQt6 for the GUI. It requires the third-party libraries **NumPy**, **Matplotlib**, **NetworkX**, and **OpenPyXL**. The source has been tested on macOS 26.1 (M4 Pro), macOS 13.6.3 (Intel Core i5), Microsoft Windows 11 Education (Update 25H2) and Ubuntu 24.04.3 LTS.

1.2.1 Run from Source

(1) Download the contents of the `src/` folder to your computer.

Example path: `/home/wangzhe/py.Aroma/src`

¹ <https://github.com/wongzit/pyNMR>

² <https://github.com/wongzit/CSigen>

(2) Ensure Python is installed (3.12 or newer recommended) and install dependencies:

```
pip install pyqt6 numpy matplotlib networkx openpyxl
```

(3) Launch the app:

```
python /home/wangzhe/py.Aroma/src/pyaroma_main.py
```

1.2.2 Use Pre-built Executable

- **macOS**

Move py.Aroma 5.app from py.Aroma-5.0-macOS-xxx-xxx.dmg to the /Applications folder. Running from other locations may cause issues.

- **Microsoft Windows**

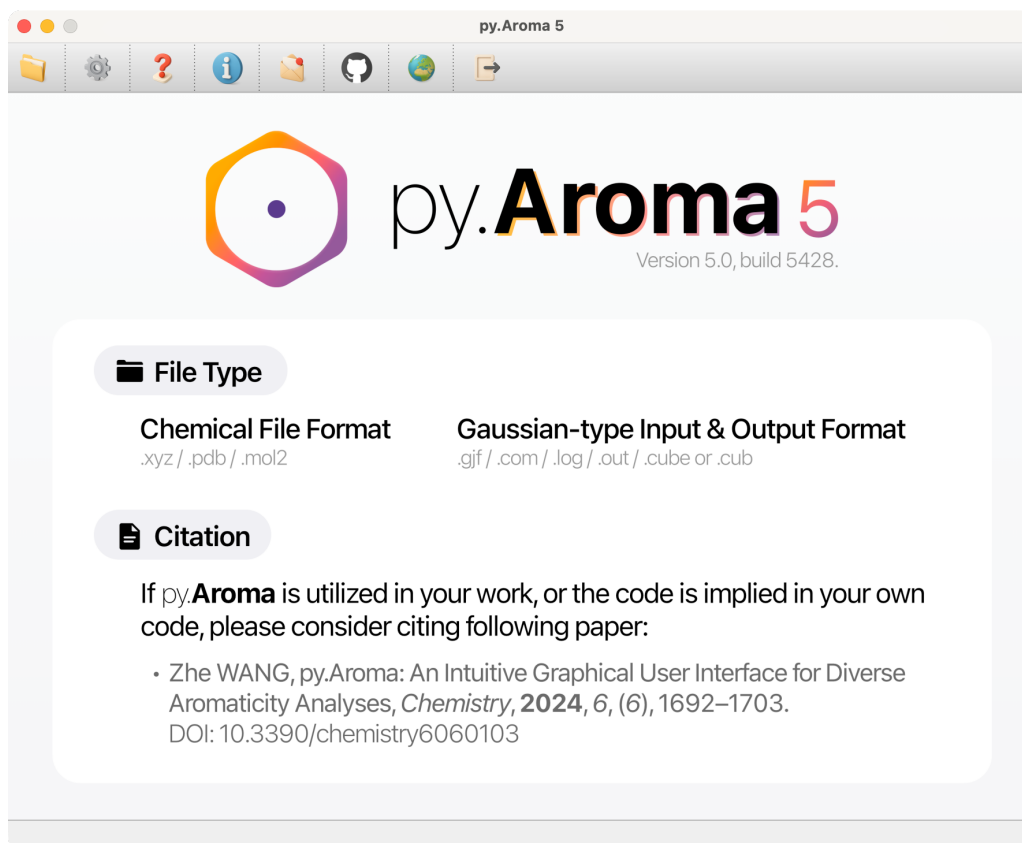
Double-click pyAroma-5.exe.

- **Linux**

From version 5, the pre-built executable will no longer be available for Linux, please use source code to build executable file using third-party library such as *PyInstaller*.

1.3 Startup

After launch, the **Start** window appears with a task bar and a short **Quick Start** guide.



 **Open** – Browse for a file to load and process in py.Aroma 5.

 **Settings** – Open py.Aroma 5 setting panel.

 **Documents** – Open the online user manual (internet required).

-  **About** – View version and license information.
-  **Contact** – Email the developer.
-  **GitHub** – Open the developer's GitHub page.
-  **Homepage** – Open the py.**Aroma** 5 homepage.
-  **Quit** – Exit the program.

2 General

This chapter covers basic information about py.Aroma 5.

2.1 File System

py.Aroma 5 read three input categories, feature availability depends on the loaded file type:

- **Type-I**

Chemical structure files: .pdb, .xyz, .mol2

Gaussian input files: .gjf, .com

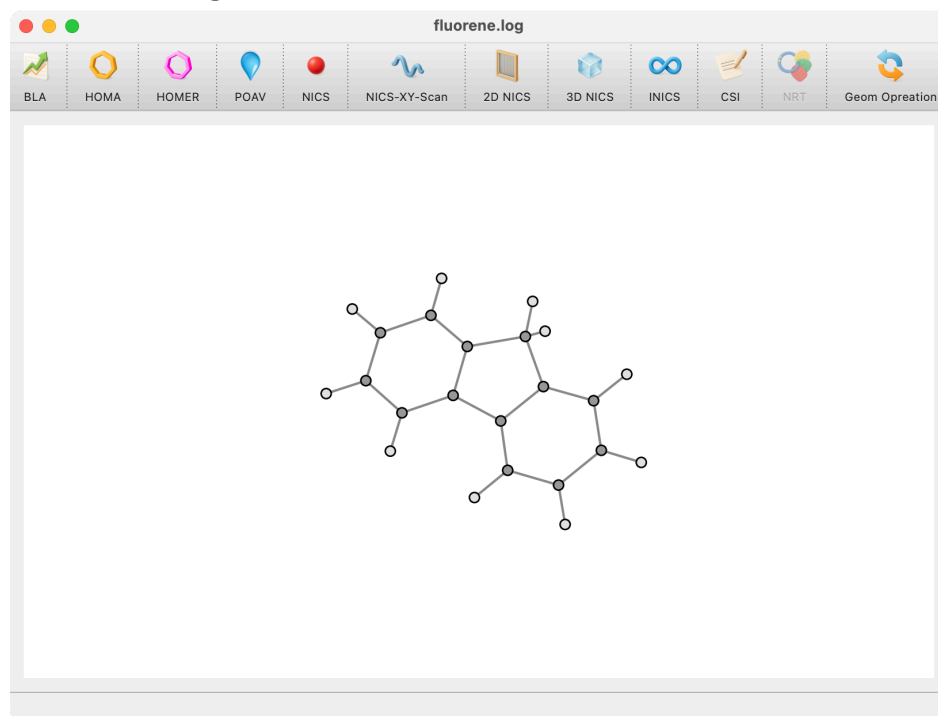
Gaussian cube files: .cube, .cube

- **Type-II**

Gaussian output files: .log or .out **without** ghost atoms (**Bq**)

- **Type-III**

Gaussian output files: .log or .out **with** ghost atoms (**Bq**)



Functionals	Icon	Type-I	Type-II	Type-III
Bond Length Alternation (BLA)		☐	☐	☐
Harmonic Oscillator Model of Aromaticity (HOMA)		☐	☐	☐
HOMER: HOMA for Excited States		☐	☐	☐
π Orbital Axis Vector (POAV)		☐	☐	☐
Create input file for single point NICS		☐	☐	–
Create input file for NICS-XY-scan		☐	☐	–
Create input file for 2D NICS/ICSS		☐	☐	–

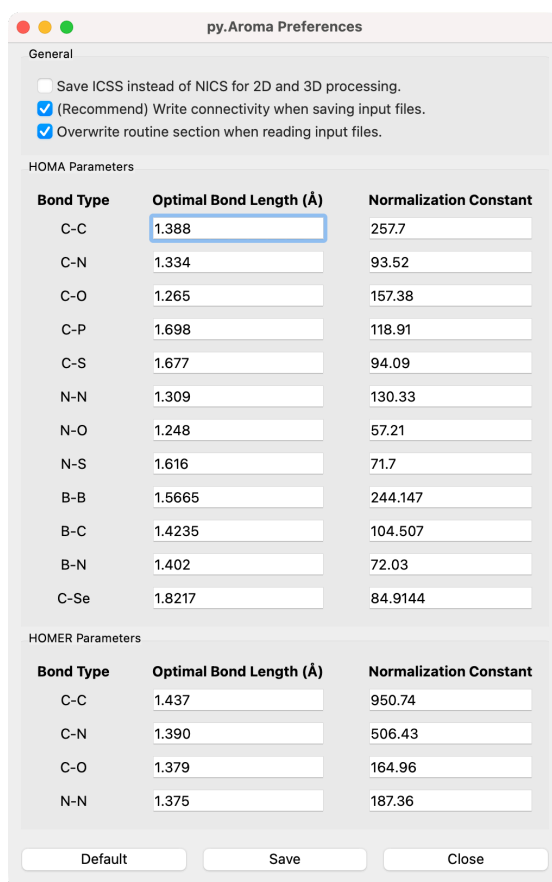
Continued Table

Functions	Icon	Type-I	Type-II	Type-III
Create input file for 3D NICS/ICSS		○	○	—
Create input file for INICS		○	○	—
Geometry Operation		○	○	—
Plot NMR Spectrum		—	Δ^a	○
Natural Resonance Theory (NRT) Analysis		—	Δ^b	—
Generate computational supporting information		—	○	—
Analyze output of single point NICS		—	—	Δ^c
Analyze output of NICS-XY-scan		—	—	Δ^c
Analyze output of 2D NICS/ICSS		—	—	Δ^c
Analyze output of 3D NICS/ICSS		—	—	Δ^c
Analyze output of INICS		—	—	Δ^c

^aNeed Gaussian output files of NMR calculation. ^bNeed NRT output file. ^cDepend on file name.

2.2 Program Setting Panel

The Settings panel provides common options and HOMA/HOMER parameters. Saved settings persist across restarts. To restore defaults, click **Default** and then **Save**.



The image shows the 'py.Aroma Preferences' dialog box. It has three tabs: 'General', 'HOMA Parameters', and 'HOMER Parameters'. The 'General' tab is selected, showing three checkboxes: 'Save ICSS instead of NICS for 2D and 3D processing.' (unchecked), '(Recommend) Write connectivity when saving input files.' (checked), and 'Overwrite routine section when reading input files.' (checked). The 'HOMA Parameters' section contains a table with columns 'Bond Type', 'Optimal Bond Length (Å)', and 'Normalization Constant'. The 'HOMER Parameters' section contains a similar table. At the bottom are 'Default', 'Save', and 'Close' buttons.

Bond Type	Optimal Bond Length (Å)	Normalization Constant
C-C	1.388	257.7
C-N	1.334	93.52
C-O	1.265	157.38
C-P	1.698	118.91
C-S	1.677	94.09
N-N	1.309	130.33
N-O	1.248	57.21
N-S	1.616	71.7
B-B	1.5665	244.147
B-C	1.4235	104.507
B-N	1.402	72.03
C-Se	1.8217	84.9144

Bond Type	Optimal Bond Length (Å)	Normalization Constant
C-C	1.437	950.74
C-N	1.390	506.43
C-O	1.379	164.96
N-N	1.375	187.36

2.3 Calculation Setup Panel

The Calculation Setup panel is available when creating a *Gaussian* input file. Click **Calculation Setup** in each module will open this panel.

In this panel, specify the NMR method, level of theory, Link 0 directives, solvation model, and other options. You may append custom keywords at the end of the input (e.g., for NBO calculations). The workflow is similar to **Calculation Setup** of *GaussView*. Click **Preview** to review the route section.

Note: If you plan to open the input files generated by py.Aroma 5 in *GaussView*, enable **Write connectivity when saving input file** in the Settings panel is highly recommended for 2D-/3D-NICS, NICS-XY-Scan and INICS (Section 2.2). This prevents *GaussView* from auto-creating bonds to ghost atoms and avoids long loading time or computer crash.

3 Functionalities

This chapter provides a detailed introduction to all the functionalities of py.Aroma 5. Functions required specific input types (**Type-I**, **Type-II** and **Type-III**), choose the proper type of files according to Section 2.1.

3.1 Bond Length Alternation (BLA)



Supported file types: **Type-I**, **Type-II** and **Type-III**

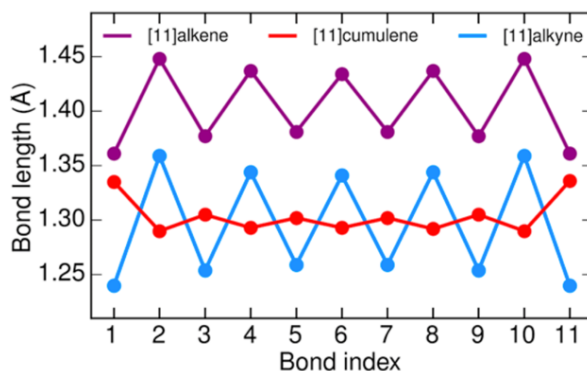
3.1.1 Theory

In conjugated systems, the bond lengths along a π -conjugated path often alternate. **Bond length alternation (BLA)** quantifies this effect and is commonly defined as the difference between the average lengths of the even and odd bonds along a specified sequence:

$$\text{BLA} = \bar{R}_{\text{even}} - \bar{R}_{\text{odd}}$$

A smaller BLA magnitude indicates stronger electron conjugation along the chosen path.

An effective method to access BLA is to plot bond length versus bond index for the specified sequence. Smaller variations across the sequence correspond to a smaller BLA value. Following Figure is an example from *J. Phys. Chem. C*, **2018**, 122, 26777:



3.1.2 Usage

To utilize this function, first define the bond sequence for the conjugated path. When you open the BLA module in py.Aroma 5, it prompts you to enter the bond indices in sequence order. The program then computes the bond lengths for those bonds, plots bond length versus bond index in the BLA window, and displays the BLA value on the plot. A color map of bond length distribution will be display in the right region of the window. You can export the plot as .png and the data as .txt or .xlsx for use in *Origin*, *GraphPad Prism*, or similar software. All outputs are saved using the input filename with the "_BLA" suffix.

An example of using BLA function in py.Aroma 5 is given in Section 4.1.2 and 4.1.3.

3.2 Calculate HOMA and HOMER Index

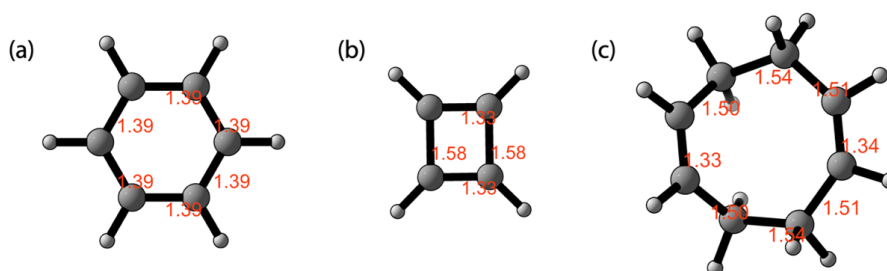
Supported file types: **Type-I**, **Type-II** and **Type-III**

3.2.1 Theory

The harmonic oscillator model of aromaticity (HOMA) is a geometry-based index widely used to evaluate aromaticity. It was introduced in *Tetrahedron Lett.*, **1972**, 13, 3839, and then the generalized form was given in *J. Chem. Inf. Comput. Sci.*, **1993**, 33, 70. The generalized HOMA is:

$$\text{HOMA} = 1 - \sum_i \frac{\alpha_{i,j}}{N} \left(R_{\text{opt}} - R_{i,j} \right)^2$$

where N is the total number of the atoms (bonds) considered, j denotes the atom next to atom i , R_{opt} and α are pre-calculated constants for each type of atom pair. If HOMA equals to 1, that means length of each bond is identical to the optimal value R_{opt} and thus the ring is fully aromatic. While if HOMA equals to 0, that means the ring is completely nonaromatic. For example, HOMA indexes of benzene, cyclobutadiene and cyclooctadiene are 0.99603 (aromatic), -3.96177 (antiaromatic) and -2.40591 (nonaromatic), respectively.



*Geometries were optimized at (R)B3LYP/6-311+G(2d,p) level of theory, orange values indicate bond lengths in Å.

In 2023, Apra and Durbeej proposed the **Harmonic Oscillator Model of Excited-state aRomaticity** (HOMER), as a supplement to HOMA, to apply this geometry-based index to excited states (*Phys. Chem. Chem. Phys.*, **2023**, 25, 16763).

The default R_{opt} and α parameters for HOMA are sourced from *Chem. Rev.*, **2014**, 114, 6383 (C-C, C-N, C-O, C-P, C-S, N-N, N-O, B-N, C-Se bonds), *Struc. Chem.*, **2012**, 23, 375 (N-S bond), *Struc. Chem.*, **2013**, 24, 543 (B-B bond), and *Struc. Chem.*, **2012**, 23, 595 (B-C bond). The HOMER parameters are sourced from *Phys. Chem. Chem. Phys.*, **2023**, 25, 16763.

Users have the option to modify these parameters from the program setting panel (Section 2.2). For unsupported bond types, you may extend functionality by adding parameters in the source code, see the detailed instructions provided in Appendix V.

3.2.2 Usage

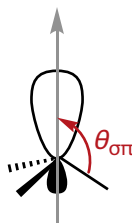
When you open the HOMA/HOMER modules, py.Aroma 5 automatically computes the HOMA/HOMER indices for all chordless monocyclic rings detected in the input geometry. To evaluate a specific ring, enter the atomic indices for that ring in connectivity order. See Section 4.2.2 for an example.

3.3 Calculate POAV Index

 Supported file types: **Type-I**, **Type-II** and **Type-III**

3.3.1 Theory

The π -orbital axis vector (POAV) analysis characterizes hybridization and out-of-plane distortion in conjugated molecules. **POAV1** theory provides $\theta_{\sigma\pi}$, θ_p ($= \theta_{\sigma\pi} - 90^\circ$), m and n values:



$$m = \frac{2 \cos^2 \theta_{\sigma\pi}}{1 - 3 \cos^2 \theta_{\sigma\pi}}$$

$$n = 3m + 2$$

where m indicates the s-orbital content ($s^m p$) of the π -orbitals and n indicates the p-orbital (sp^n) content of the σ -orbitals.

POAV2 provides $\theta_{1\pi}$, $\theta_{2\pi}$, $\theta_{3\pi}$, n_1 , n_2 , n_3 and m . More details about the POAV1 and POAV2 theory can be found in *J. Am. Chem. Soc.*, **1986**, 108, 2837 and *J. Phys. Chem. A*, **2001**, 105, 4164. The one-center POAV1 and POAV2 function in py.Aroma 5 is based on the three sigma bond angles. For an ideal sp^2 hybridized carbon atom, $\theta_{1\pi} = \theta_{2\pi} = \theta_{3\pi} = 120^\circ$, $\theta_{\sigma\pi} = 90^\circ$, $\theta_p = 0^\circ$, $m = 0$ and $n = n_1 = n_2 = n_3 = 2$.

3.3.2 Usage

By entering the index of the target atom, py.Aroma 5 computes and displays the POAV1 and POAV2 parameters in the results window.

It's important to note that the inputted atom **MUST** have and **ONLY** have three bonded neighboring atoms. An example of using POAV function in py.Aroma 5 is given in Section 4.3.2.

3.4 NICS Calculation

3.4.1 Theory

3.4.1.1 Single Point NICS

The nucleus-independent chemical shift (NICS) is a widely used index for evaluating aromaticity through theoretical computation. A negative NICS index indicates an aromatic system, while a positive index indicates an antiaromatic system. A NICS index close to 0 signifies nonaromaticity or weak aromaticity/antiaromaticity. Several studies, such as the one published in *Org. Lett.*, **2006**, 8, 863, have shown that $\text{NICS}(0)_{zz}$ or $\text{NICS}(1)_{zz}$ is a better index than the original definition of NICS, now known as $\text{NICS}(0)$.

The most common method to obtain NICS values is to add ghost atoms (in *Gaussian*: **Bq** atom) at the target positions and then submit to NMR computation. The shielding tensors of all atoms are computed and summarized in the output file. A typical output for the shielding tensors of ghost atoms is shown below:

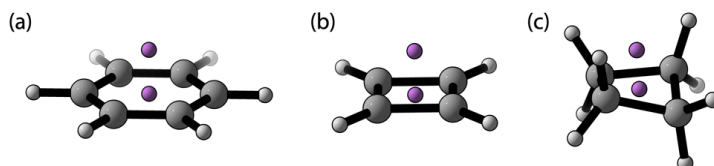
```

          9  Bq  Isotropic =   -0.1133  Anisotropy =    1.6782
XX=      0.0988  YX=      0.5330  ZX=      0.3438
XY=      0.5353  YY=      0.1032  ZY=      0.3474
XZ=      0.7234  YZ=      0.7264  ZZ=     -0.5420

```

The NICS index represents the reversed value of the shielding tensor. In the example above, NICS is +0.1133 (indicated by "Isotropic =") and NICS_{zz} is +0.5420 (indicated by "ZZ=").

For instance, NICS indexes calculated at (R)B3LYP/6-311+G(2d,p) for benzene, cyclobutadiene, and cyclobutane are shown:



Molecule	NICS(0)	NICS(1)	NICS(0) _{zz}	NICS(1) _{zz}
benzene	-7.61	-10.00	-14.97	-29.68
cyclobutadiene	+27.05	+17.72	+110.79	+55.62
cyclobutane	+0.38	+1.24	+56.76	+3.14

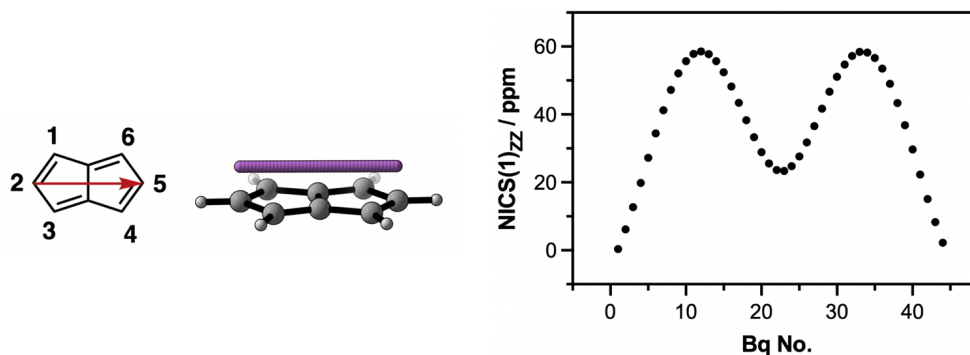
Dissected NICS (e.g., $\text{NICS}(1)_{\pi,zz}$) can offer sharper insight into aromaticity (see *Org. Lett.*, **2006**, 8, 863). Dissection may be carried out with CMO-based (via NBO program) or σ -only models. Appendix VI provides a step-by-step guide for CSGT-based calculations.

3.4.1.2 NICS-XY-Scan

NICS is often studied at specific points, such as the center of a ring. To provide a more detailed understanding of the induced magnetic field of a molecule, a method called NICS-XY-Scan (See: *Chem. Eur. J.*, **2014**, 20, 5673) has been proposed, which involves using a series of NICS probes,

where the molecule is positioned on the XY plane, the trajectory of NICS probes is parallel to the XY plane, and the probes are placed at 0.1 Å intervals along a line that spans the length of the system.

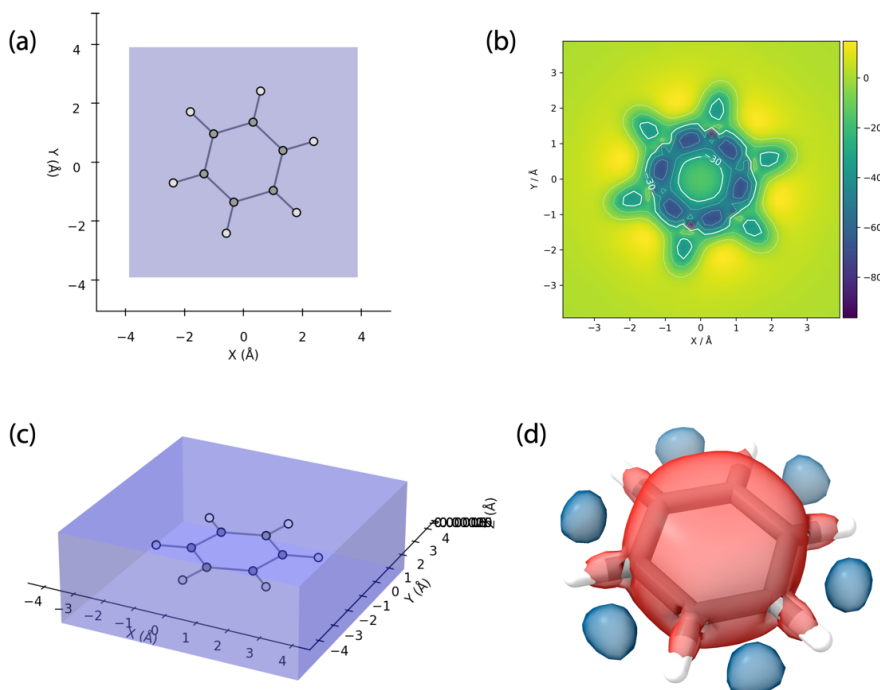
An example of a NICS-XY-Scan of pentalene, calculated at the (R)B3LYP/6-311+G(2d,p) level of theory, is illustrated. In this example, the NICS probes are aligned along the C2 to C5 with 0.1 Å intervals and are positioned 1 Å above the pentalene plane.



3.4.1.3 2D and 3D NICS Analysis

Like the XY-scan, 2D and 3D NICS provide spatial maps of magnetic shielding. In some contexts these are described as 2D/3D ICSS (iso-chemical shielding surfaces), which use raw shielding tensors instead of the sign-reversed NICS values. Thus 2D/3D ICSS and 2D/3D NICS derive from the same data and have equal magnitudes with opposite signs.

Examples of 2D and 3D NICS analyses of benzene are depicted with the NMR computations performed at the (R)B3LYP/6-311+G(2d,p) level of theory.



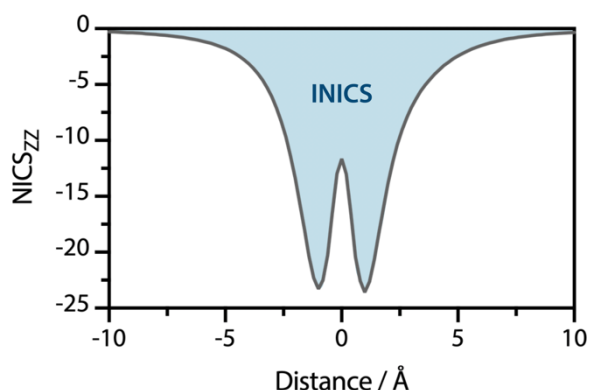
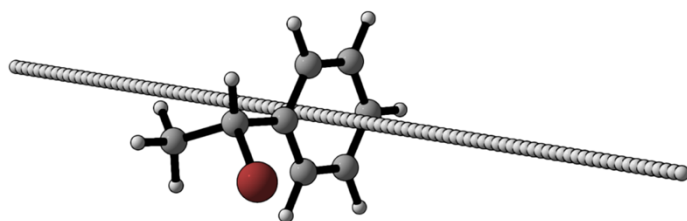
*Illustrates of (a) 2D NICS and (c) 3D NICS, the violet regions indicate the positions of ghost atom. (b) give a 2D NICS(0)_{zz} heatmap and the contour line of -30 ppm is emphasized in bold line. (d) show the iso-surface of -25 ppm (red) and 10 ppm (blue) from the data of 3D NICS_{zz}.

3.4.1.4 Integral NICS

Integral NICS (**INICS**) evaluates aromaticity in **asymmetric** environments where, i.e.: $\text{NICS}(l) \neq \text{NICS}(-l)$ (*J. Phys. Chem. A*, **2022**, 126, 3433). The INICS index is defined as the area of NICS trace follows:

$$\text{INICS}_\alpha = \int_{-l}^l \text{NICS}_\alpha(z) dz$$

For example, (1-bromoethyl)benzene has different chemical environment on the two sides of phenyl ring, thus, $\text{NICS}(0)_{zz} = -9.40$, $\text{NICS}(1, -\text{CH}_3)_{zz} = -14.22$, $\text{NICS}(-1, \text{Br})_{zz} = -14.52$, $\text{INICS}(-10, 10)_{zz} = -65.61$. The range and interval of NICS probes make influence on the INICS index (Appendix III). The default parameters (from 10 Å below the ring plane to 10 Å above the ring plane, with 0.2 Å interval) are enough for most cases.



3.4.2 Create Input File for NICS Calculation

 Supported file types: **Type-I** and **Type-II**

Users can generate *Gaussian* input files for NICS calculations using py.Aroma 5. Upon opening the **Type-I** or **Type-II** input file, users can select **NICS**, **NICS-XY-Scan**, **2D NICS**, **3D NICS**, or **INICS** based on their specific requirements.

In the input file creation window, users have the option to edit the Gaussian calculation keyword via the **Calculation Setup** button. However, when reading a .gjf or .com file as a py.Aroma 5 input, the **Calculation Setup** is only applicable when the **Overwrite routine section when reading input files** option is enabled in the program setting window (Section 2.2). This option is checked by default.

3.4.2.1 Single Point NICS

Upon entering the single point NICS window, py.Aroma 5 will automatically detect all chordless monocycles in the geometry, allowing users to add ghost atoms for all monocycles at once. If a sequence of atom indices (requiring more than three atoms) is provided, the ghost atom will be added at the center of the specified atoms. Users also have the option to adjust the height. If a height value greater than 0 is specified, py.Aroma 5 adds two ghost atoms above and below the plane. py.Aroma 5 also allows users to add ghost atom at center of geometry or center of mass.

The .gjf files for NICS calculations will be saved with the same file name as the input file, followed

by the “_NICS” suffix.

For an example of creating an input file for single point NICS calculation with py.Aroma 5, please refer to Section 4.4.2.

3.4.2.2 NICS-XY-Scan

To create a path for a NICS-XY-scan, a sequence of knots (at least two knots) is required. The knots will be added at the center of the specified atoms. Once more than two knots are specified, the path will be displayed on the screen, and the figure can be saved as a .png image with the same file name as the input file, followed by the “NICS_Scan” suffix. py.Aroma 5 also supports NICS-XY-scan on the YZ and XZ planes, which can be selected in the parameter section, including intervals and height above the plane. The input file will be saved with the same name as the .png file. An example of creating an input file for a NICS-XY-scan calculation with py.Aroma 5 is provided in Section 4.5.2.

3.4.2.3 2D and 3D NICS

Creating the input files for 2D/3D NICS calculation with py.Aroma 5 is straightforward. Users only need to specify the region for adding ghost atoms, which can be checked by clicking the **Preview** button. The .gjf files for 2D/3D NICS calculation will be saved with the same file name as the read file, followed by the “_2D(3D)_NICS” suffix. Examples of creating input files for 2D NICS calculation with py.Aroma 5 are provided in Section 4.6.2 and see Section 4.7 for 3D NICS.

3.4.2.4 INICS

INICS requires adding NICS probe from distance l below the ring plane to distance l above the ring plane. py.Aroma 5 provides a simple protocol to create input files for INICS calculations via *Gaussian*. Users need to specify the range l and interval j of NICS probe path. The ghost atoms will be added in the range from $-l$ Å to l Å, in every j Å. Same as NICS module, the chordless monocycles could be automatically detected, thus, users can add NICS probe for all monocycles at once. Also, users can specify the cyclic structure by inputting atom indexes.

The NICS probes will be updated automatically in the preview window. Users can easily confirm the position of ghost atoms. The .gjf files will be saved with the same file name as the file read, with “_INICS” suffix. An example of creating an input file for a INICS calculation with py.Aroma 5 is provided in Section 4.8.2.

3.4.3 Process Output File of NICS Calculation

 Supported file types: **Type-III**

Users can process output files of NICS calculations with py.Aroma 5. For analyzing output files of NICS scan, 2D NICS, 3D NICS and INICS, the output files must be generated by the input files created by py.Aroma 5. For 2D and 3D NICS, by default, the NICS indices (i.e., reversed values of

shielding tensors) will be used. If 2D and 3D ICSS is preferred, please check the **Save ICSS instead of NICS for 2D and 3D processing** option in the program setting panel (Section 2.2).

3.4.3.1 Single Point NICS

py.Aroma 5 automatically finds the shielding tensors from the output file and display the NICS values on the screen. Users can choose the component of the shielding tensors. The results can be saved as a .txt file with the same file name as the *Gaussian* output, followed by the "_NICS_output" suffix. For those rings which are not parallel to the Cartesian planes (i.e., XY, YZ, XZ), py.Aroma 5 can compute pseudo-NICS_{zz}, i.e., NICS_⊥, for those rings, based on the shielding tensors and normal vector of fitted plane. An example can be found in Section 4.9.2. For details about NICS_⊥, please see *Chemistry*, **2024**, 6, 1692.

3.4.3.2 NICS-XY-Scan

py.Aroma 5 will automatically find the shielding tensors from the output file, display the NICS values, and plot the NICS values versus indices of ghost atoms on the screen. Users can also choose the component of the shielding tensors, and the results will be updated automatically. The results can be saved as a .png and .xlsx with the "_NICS_Scan_output" suffix. An example can be found in Section 4.9.3.

3.4.3.3 2D NICS

When reading output files for 2D NICS, the heatmap will be automatically displayed on the screen. Users can choose the component of the shielding tensors, and the heatmap will be updated automatically. The heatmap can be saved as a .png image with the "_2D_NICS_output" or "_2D_ICSS_output" suffix in the file name. The NICS (or ICSS) indexes can also be saved as a .csv file for further investigation. An example can be found in Section 4.9.4.

3.4.3.4 3D NICS

After entering 3D NICS window, user could choose the component of shielding tensors and save them to a *Gaussian* .cube (same as .cub) file, with "_3D_NICS_output" or "_3D_ICSS_output" suffix in file name, which could be visualized by *GaussView*, *VMD*, *ChimeraX*, etc. A preview of the heatmap for a 2D slice will be displayed on the screen. Users can edit it in the same way as 2D NICS module. An example can be found in Section 4.9.5.

3.4.3.5 INICS

The INICS window displays the geometry and the NICS profile. If multiple rings are present, Ring No. 1 is shown by default; choose the ring and tensor component of interest. The computed INICS value appears in the right panel. Save the plot as .png via **Save .png** and the values as .xlsx via **Save .xlsx**. See Section 4.9.6.

3.5 NMR Analysis

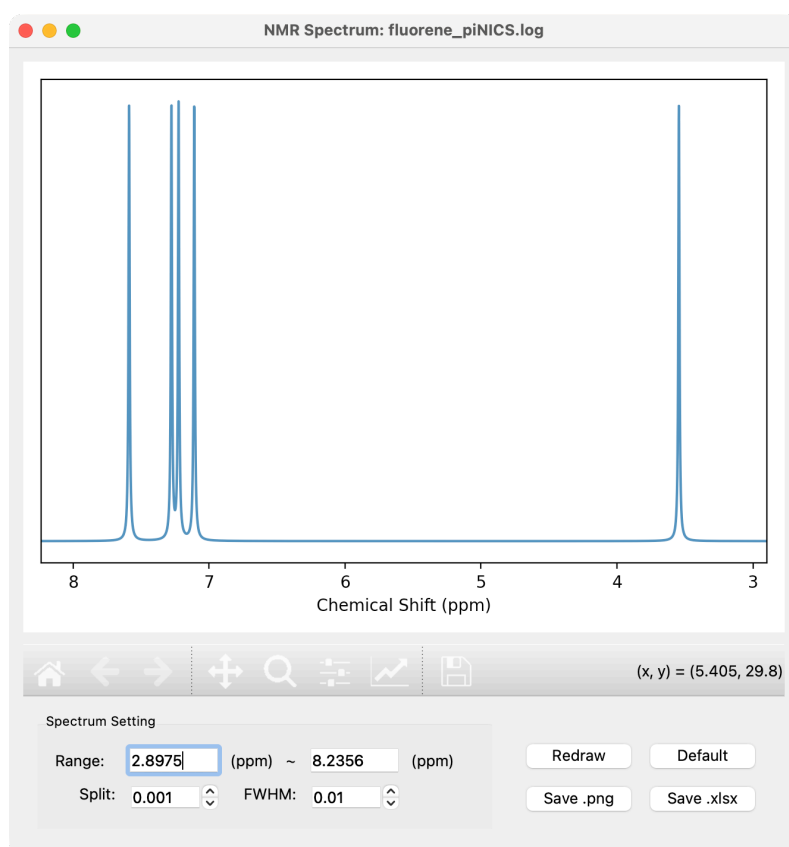
Supported file types: **Type-II** (NMR output only) and **Type-III**

The py.NMR tool for applying scaling factors to computed NMR chemical shifts is integrated into py.Aroma 5 with expanded capabilities. When an NMR output file is loaded, the NMR module becomes available.

In the NMR window, select the element (nucleus) to plot. Three modes are provided:

1. **Isotropic shielding** — plot values read directly from the output.
2. **Referenced shifts** — set a reference and plot chemical shifts relative to it.
3. **Scaled shifts** — apply a user-defined scaling factor to the computed shifts.

See Section 4.10 for an example workflow.



Note: py.NMR is no longer being maintained by the developer since the basic function has been combined in py.Aroma from version 4.0, but it can still be downloaded from the archived GitHub page (<https://github.com/wongzit/pyNMR>).

3.6 Generate Computational Supporting Material

Supported file types: **Type-II**

The open-source CSIgen script for generating computational supporting information is integrated into py.Aroma 5. When a Type-II file is loaded, the CSIgen tool becomes available.

fluorene							
				#p opt freq rhf/sto-3g			
				Charge = 0, Multiplicity = 1, Point group = C1			
				Number of imaginary frequencies = 0			
				Electronic Energy = -492.09227558 Hartree			
				Sum of electronic and zero-point Energies = -491.873787 Hartree			
				Sum of electronic and thermal Energies = -491.866015 Hartree			
				Sum of electronic and thermal Enthalpies = -491.865071 Hartree			
				Sum of electronic and thermal Free Energies = -491.907143 Hartree			
Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
H	-0.000033	2.483012	-0.879198	C	3.009779	-1.203774	-0.000148
C	3.449819	0.111756	-0.000139	C	2.540004	1.164723	-0.000082
C	1.189207	0.877596	-0.000032	C	0.744188	-0.449011	-0.000038
C	1.651439	-1.496204	-0.000089	H	3.732059	-2.010202	-0.000193
H	4.512030	0.321418	-0.000185	H	2.892120	2.188584	-0.000081
H	1.314262	-2.524941	-0.000096	C	0.000000	1.837208	0.000026
H	0.000034	2.482988	0.879268	C	-1.189207	0.877596	0.000057
C	-2.540004	1.164723	0.000108	C	-3.449819	0.111756	0.000162
C	-3.009780	-1.203774	0.000094	C	-1.651439	-1.496204	0.000054
C	-0.744189	-0.449011	0.000024	H	-2.892120	2.188584	0.000134
H	-4.512030	0.321418	0.000207	H	-3.732059	-2.010202	0.000120
H	-1.314262	-2.524941	0.000036				

In the CSIgen window, choose the items to include and export the supporting information as .txt or .xlsx. See Section 4.11 for an example.

Note: CSIgen is no longer being maintained by the developer since the basic function has been combined in py.Aroma from version 3.0, but it can still be downloaded from the archived GitHub page (<https://github.com/wongzit/CSIgen>).

3.7 Other Functions

3.7.1 Natural Resonance Theory (NRT) Analysis



Supported file types: **Type-II** (require NRT output section)

py.Aroma 5 includes an NRT module for analyzing Natural Resonance Theory results from NBO calculations. When an output containing NRT data is loaded, the tool becomes available. See Section 4.12 for an example.

3.7.2 Geometry Operation



Supported file types: **Type-I** and **Type-II**

py.Aroma 5 provides a Geometry Operations module for editing structures and saving the modified geometry. See Section 4.13 for an example.

4 Tutorials and Examples

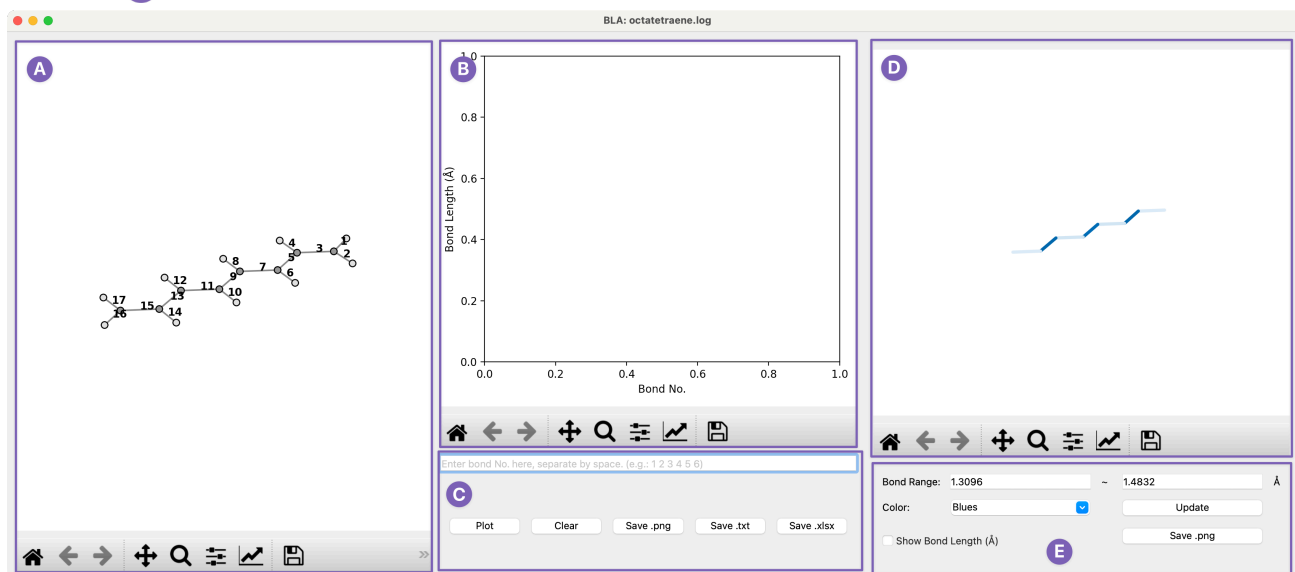
This chapter includes detail tutorials and some examples for using py.Aroma 5 to carry out aromaticity analyses.

4.1 BLA Analysis

4.1.1 BLA Interface

Open a file, then click **BLA**. The window contains five areas:

- A Geometry view** — displays the molecular structure with bond indices at each bond center.
- B Plot preview** — shows the bond-length vs. bond-index graph. No preview appears until you click **Plot** in **C**.
- C Controls** — enter bond indices in sequence; buttons: **Plot** (update **B** and compute BLA), **Clear** (clear **B**), **Save .png** (export plot in **B**), **Save .txt** / **Save .xlsx** (export bond data in **B**).
- D Bond length distribution map** — color map of bond length distribution.
- E Map options** — set bond length range and color scale, toggle value labels, and save the map in **D**.



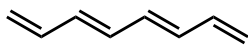
To plot the BLA graph, enter a space-separated sequence of bond indices in the text field in **C**, for example: 3 5 7 9 11 13 15. Use spaces to separate index only. Do not use commas, periods, letters, or any other separators. Indices must be integers and must correspond to existing bonds in the current molecule.

Invalid examples:

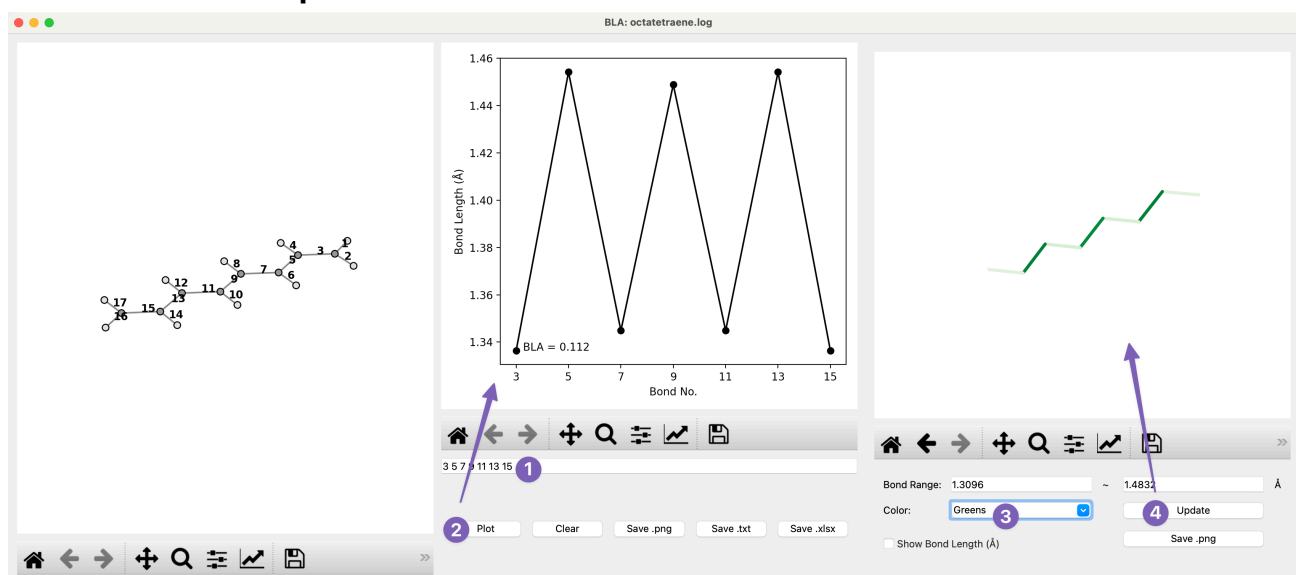
- 3,5,7,9,11,13,15 — illegal separator (commas)
- 3.5.7.9.11.13.15 — illegal separator (periods)
- 3 5 7 9 11 13 22 — nonexistent bond index (22)
- 3 5 7 9 11 13 1t — non-numeric token (t)

4.1.2 BLA Analysis of 1,3,5,7-Octatetraene

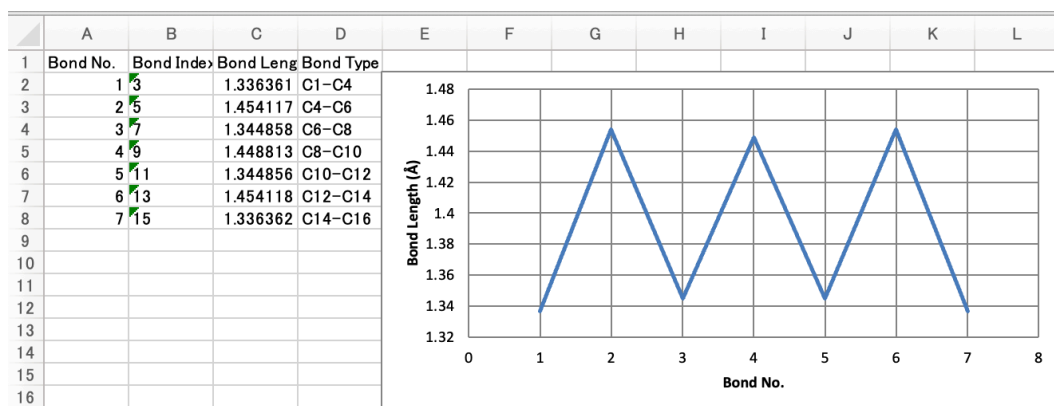
Structure optimization of 1,3,5,7-octatetraene was performed at (R) ω B97X-D/6-31G(d) level of theory. The output file octatetraene.log was read by py.Aroma 5.



Enter the bond sequence 3 5 7 9 11 13 15 in the text field in **C** then click **Plot** to display the BLA preview in **B**. To adjust the bond-length color map, change the scale from **Blues** (default) to **Greens** and click **Update**.



To save results, use **Save .png**, **Save .txt**, or **Save .xlsx** in **C**. Files are written as octatetraene_BLA.png, octatetraene_BLA.txt, and octatetraene_BLA.xlsx. An example .xlsx output is shown below.

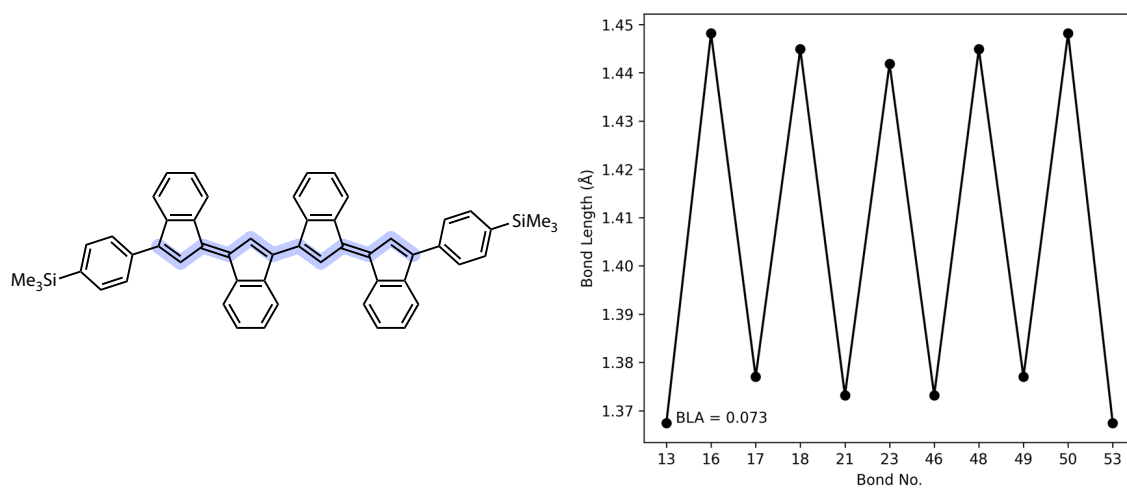


Hint: After entering a valid bond sequence, you can save directly with **Save .png**, **Save .txt**, or **Save .xlsx** without clicking **Plot**; the preview module runs automatically before export.

4.1.3 BLA Analysis of Oligo(biindenylidene)

Follows example is from *Nat. Commun.*, **2023**, 14, 2741. The structural coordinates were extracted from supporting information, saved as natcom_4b.xyz, and loaded into py.Aroma 5. Enter the bond sequence 13 16 17 18 21 23 46 48 49 50 53 and plot the BLA graph. As

shown below, the resulting plot closely reproduces Figure 4b from the paper by py.Aroma 5.

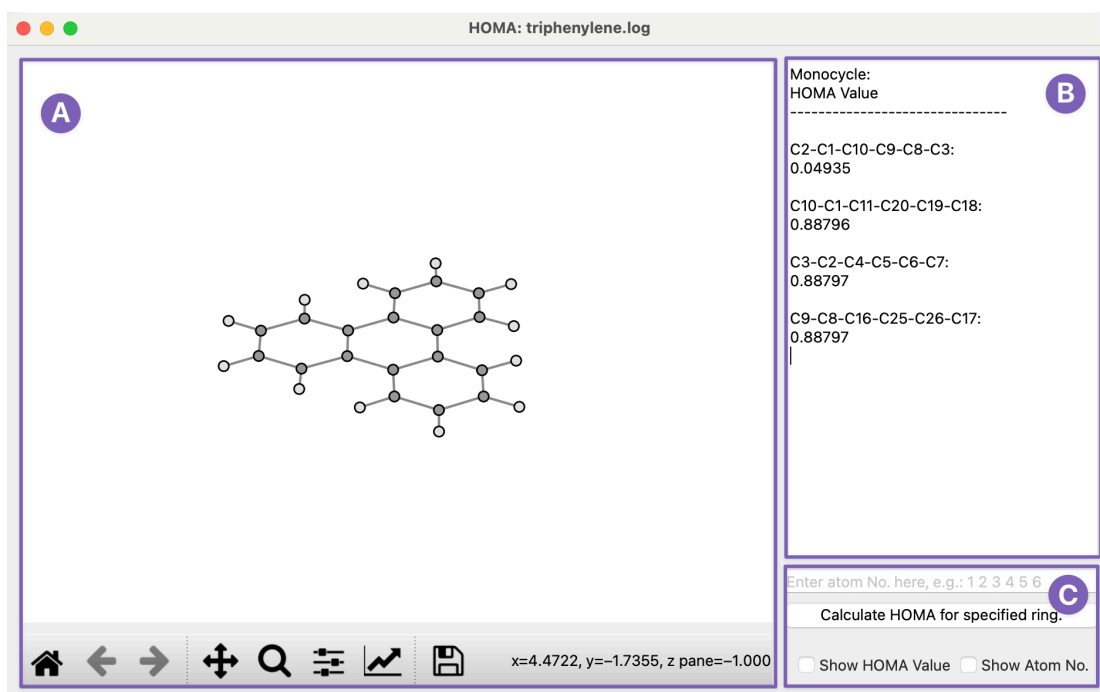


4.2 HOMA and HOMER Analysis

4.2.1 HOMA and HOMER Interface

The HOMA and HOMER functions can be accessed by clicking the HOMA/HOMER button after opening a file.

- A Geometry view** — displays the molecular structure.
- B Results** — lists the atomic indices for each detected monocycle and its HOMA/HOMER value.
- C User input** — enter a sequence of atomic indices for a specific ring in connectivity order, then click **Calculate HOMA/HOMER** for specified ring. The check boxes **Show HOMA/HOMER Value** can show/hide the HOMA/HOMER values at the center of each cycle in **A**; **Show Atom No.** can show/hide atom indices on each atom in **A**, which is useful when specifying rings in the text bar.



The R_{opt} and α parameters used for HOMA/HOMER calculations are defined in program setting panel. User can modify the parameters if necessary. More information about program setting, please refer to Section 2.2.

Example format in **B**:

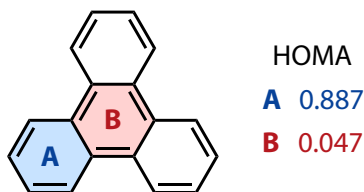
C10-C18-C19-C20-C11-C1:
0.88796

which indicates a HOMA/HOMER value of ring **C10-C18-C19-C20-C11-C1** is **0.88796**.

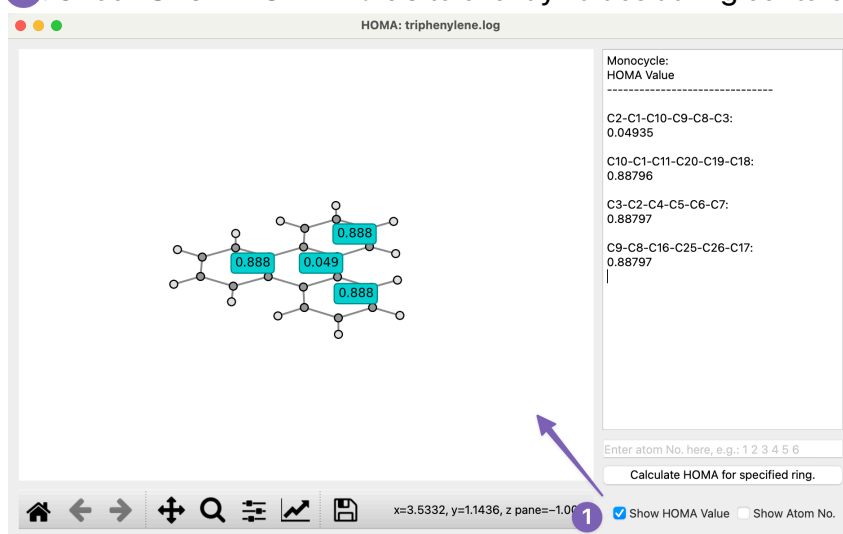
Same as the BLA function, the atom indexes in the text bar of **C** should also be separated by spaces, e.g.: 10 18 19 20 11 1, and the order should be the same as bond connectivity. For example, 10 19 18 20 11 1 is not valid since there is no connection between 10 and 19, 18 and 20. The sequence MUST include more than three atom indexes to construct a cyclic structure.

4.2.2 HOMA Analysis of Triphenylene

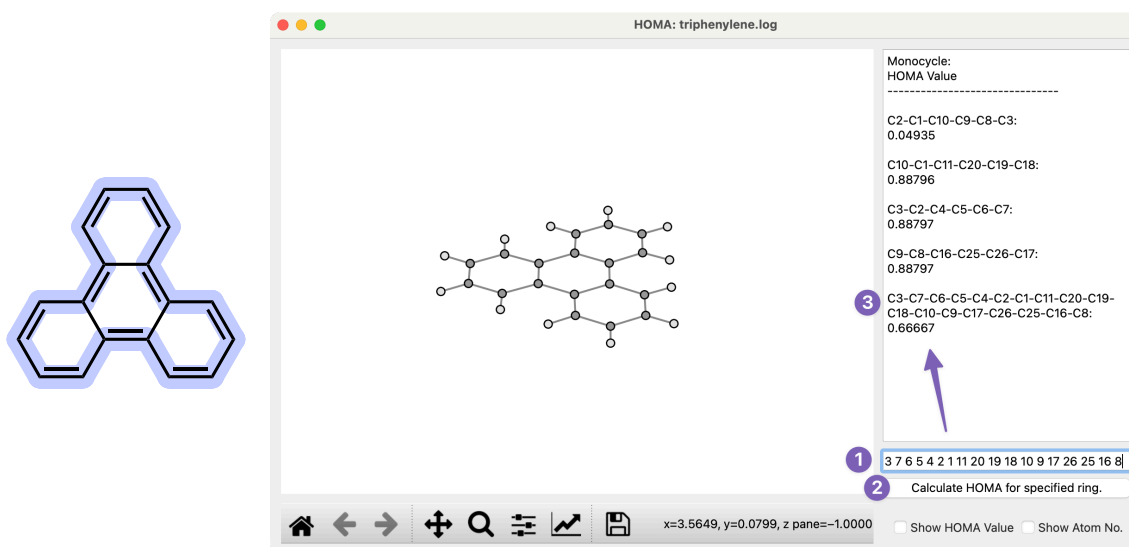
Structure optimization of triphenylene was performed at (R)B3LYP/6-31G(d) level of theory. The optimized triphenylene adapted D_{3h} point group and the HOMA values of ring A and ring B have been reported to be 0.887 and 0.047, respectively, from *Org. Lett.*, **2014**, 16, 6116.



Load the optimized geometry in py.Aroma 5. HOMA values for all detected monocycles appear automatically in **B**. Check **Show HOMA Value** to overlay values at ring centers in **A**.



To evaluate the outer rim, enter the atom sequence, a sequence of atom indices (3 7 6 5 4 2 1 11 20 19 18 10 9 17 26 25 16 8) is entered into the text bar, followed by clicking the **Calculate HOMA for specified ring** button. Subsequently, the HOMA value will be displayed in **A**.

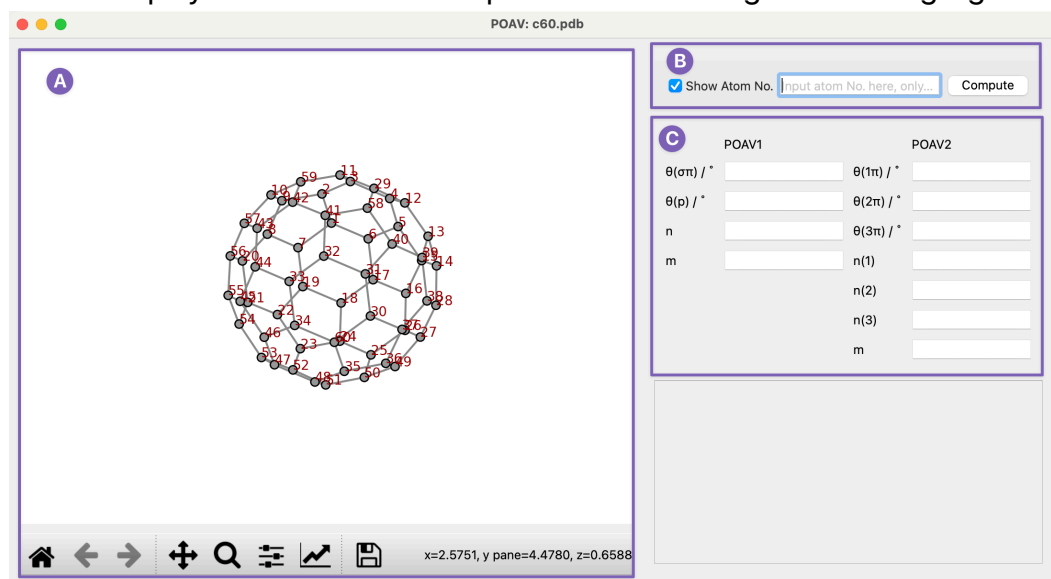


4.3 POAV Analysis

4.3.1 POVA Interface

Users can access the POAV function by clicking the POAV button after loading a file.

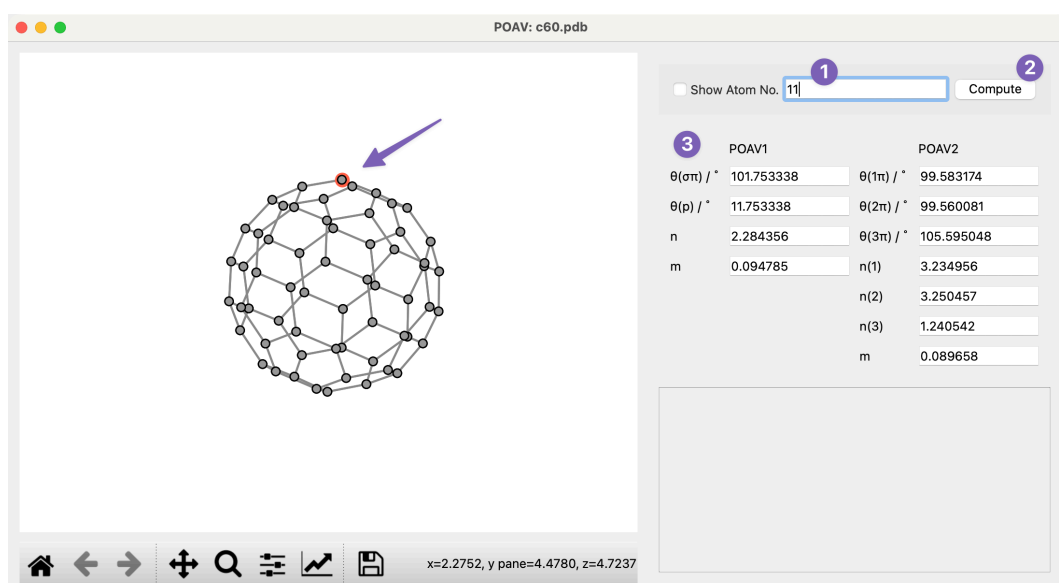
- A Geometry view** — displays the molecular structure.
- B Controls** — **Show Atom No.** checkbox toggles atom indices in **A**. Text field for inputting the target atom index. The **Compute** button runs the analysis.
- C Results** — displays POAV1 and POAV2 parameters. The target atom is highlighted in **A**.



The text field in **B** accepts one atom index only. The target atom must have exactly three bonded neighbors.

4.3.2 POAV Analysis of Fullerene C₆₀

Using fullerene C₆₀ as example, to obtain POAV parameters for atom C11, enter **11** in the text field and click **Compute**. Atom C11 is highlighted and the POAV1/POAV2 results appear in **C**.



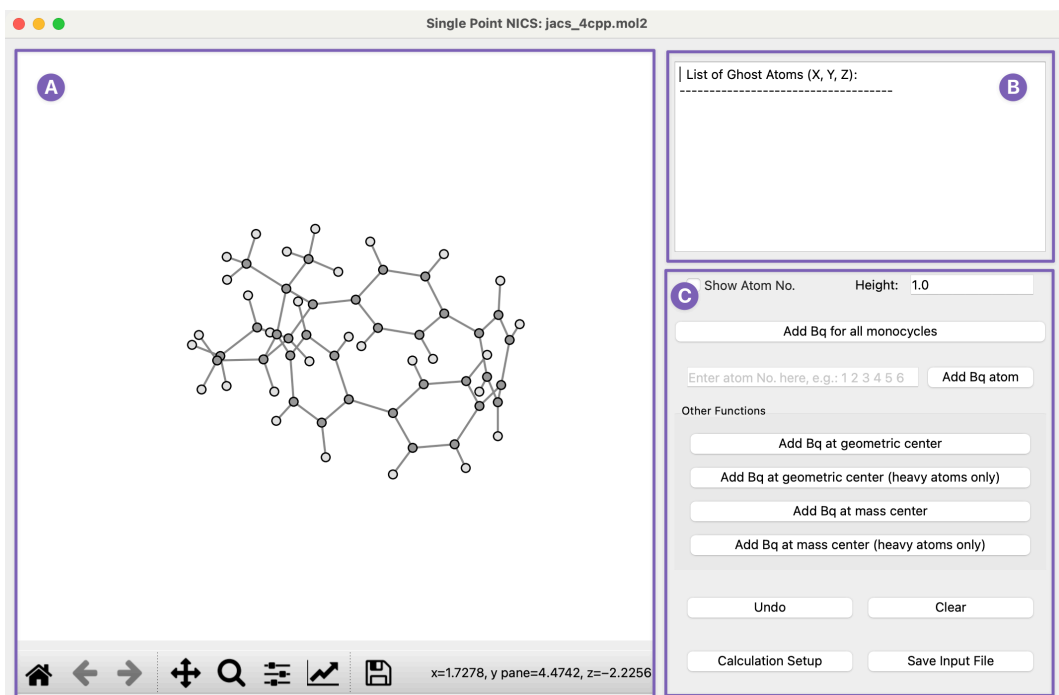
4.4 Create Input File for NICS ●

4.4.1 Basic Information

The single point NICS function can be accessed by clicking the **NICS** button after loading a file.

- A Geometry view** — displays the molecular structure.
- B Coordinates** — displays the coordinates of ghost atoms.
- C Control** — By clicking the **Add Bq atoms for all monocycles** button, py.Aroma 5 will add ghost atoms at a specified height for all monocycles (1 Å by default, but the height can be modified in the **Height** text field). Furthermore, users can customize the position of the ghost atom by follows these steps:

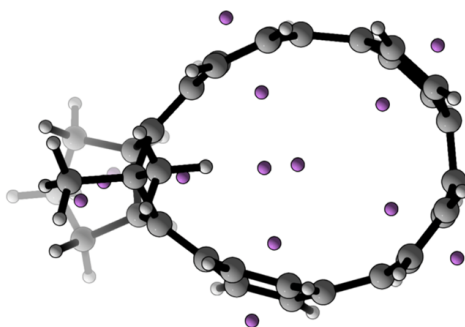
- (0) Check **Show Atom No.**, although this is not compulsory, it is convenient.
- (1) Input the atom indexes in the text field (e.g.: 1 2 3 4 5 6).
- (2) Set the height in the **Height** text field.
- (3) Click the **Add Bq atom** button.



py.Aroma 5 ignores monocycles larger than 10 members. For larger rings, add ghost atoms by specifying atom indices in **C**. Use **Undo** to remove the most recently added ghost atom and **Clear** to remove all ghost atoms. The Cartesian coordinates of ghost atoms are listed in **B**. To specify a custom position, provide ≥ 3 atom indices to define a plane in **C**. Order is insensitive (e.g., 1 2 3 4 5 6 and 1 5 3 6 4 2 are equivalent). If Height $\neq 0$ (i.e., NICS(l) with $l > 0$), two ghost atoms are placed above and below the plane. Users can also place a ghost atom at the center of geometry or center of mass, using all atoms or heavy atoms only; these options are available in the button group in **C**. Click **Save Input File** to write the *Gaussian* input. Adjust computation details via **Calculation Setup** before saving.

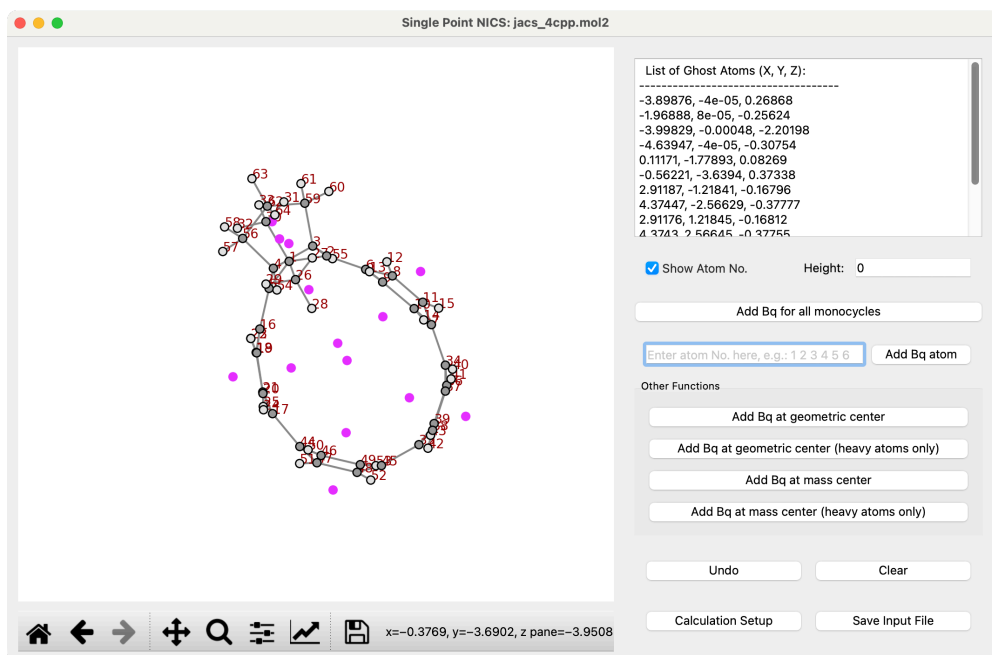
4.4.2 Example: DR-[4]CPP

To give a better understanding of how to use NICS module in py.Aroma 5, a step-by-step procedure using **DR-4CPP** (*J. Am. Chem. Soc.*, **2021**, *143*, 7426.) would be described. The target is: (a) NICS(1) for all monocyclic rings; (b) NICS(0) at the center of macrocycle; (c) NICS at center of mass (heavy atom only).



< Procedure >

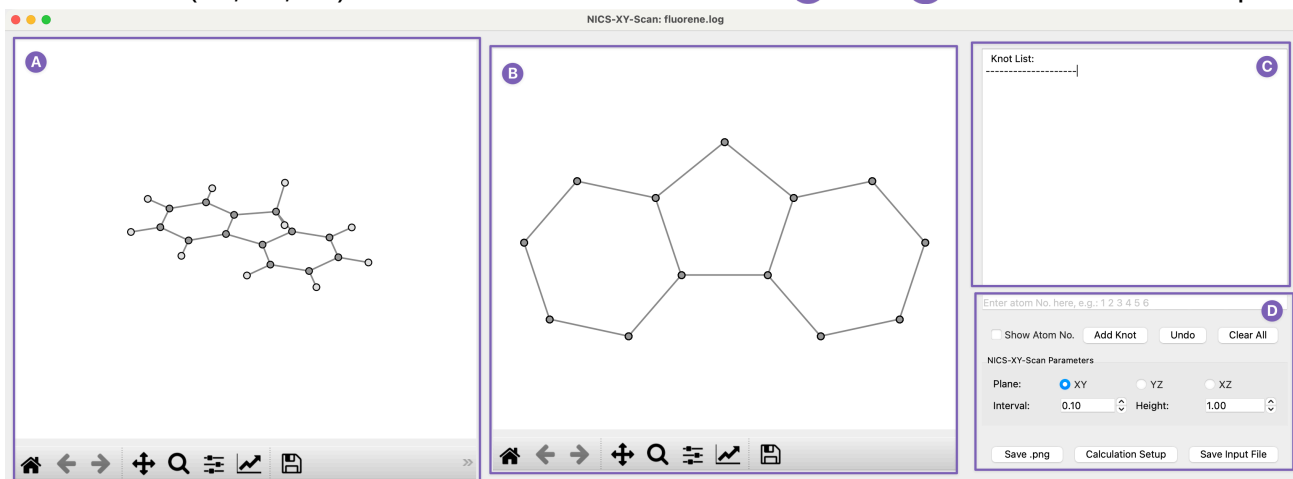
- (0) Open `jacs_4cpp.mol2` in py.Aroma 5.
- (1) Click **Add Bq atoms for all monocycles**, ghost atoms for NICS(1) will be added. Target (a) done.
- (2) Check **Show Atom No..**
- (3) Change the **Height** value to 0.
- (4) Input the atom sequence in the text field: 2 5 6 7 34 35 44 45 16 17. In this example, the equatorial atoms are used to define the plane.
- (5) Click **Add Bq Atom**, the NICS(0) of the macrocycle will be added. Target (b) done.
- (6) Click **Add Bq atoms at mass center (heavy atom only)**. Target (c) done.
- (7) Click **Save Input File**, finished!



4.5 Create Input File for NICS-XY-Scan

4.5.1 Basic Information

Click **NICS-XY-Scan** after loading a file. The 3D molecular geometry appears in **A**, and its projection on the XY plane appears by default in **B**. To change the projection plane, use the **Plane** radio buttons (XY, YZ, XZ) in NICS-XY-Scan Parameters in **D**. The **C** list the knots of scan path.



A path is required for the NICS-XY-scan calculation. Create the path by adding knots according to following steps:

- (0) Check **Show Atom No.**, which is not compulsory but convenient.
- (1) Input the atom indexes in the text field in **D** (e.g.: 1 2 3 4 5 6).
- (2) Click **Add Knot**, and the knot will be added at the center of the inputted atoms. The knot information will be updated in **C**. (**Note:** If users inputted one atom index, the atom will be the knot.)
- (3) Repeat steps (1) and (2) until the full path is finished.

Manage knots with **Clear All** (remove all) or **Undo** (remove last). To generate a *Gaussian* input, click **Save Input File**. py.Aroma 5 places ghost atoms along the knot path at the specified **Height** (default 1 Å, editable in **D**) with a probe **Interval** (default 0.1 Å, editable). Save the projection figure in **B** via **Save .png** in **D**.

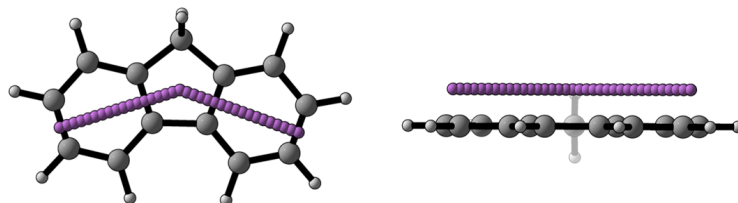
4.5.2 Example: Fluorene

Follows is a step-by-step tutorial for how to use NICS-XY-scan module in py.Aroma 5, with fluorene as example. The fluorene molecule is placed on XY plane, we aim to put the NICS probes on the trajectory as shown below, on the height (Z axis) of 1 Å, with 0.2 Å interval of ghost atoms.

< Procedure >

- (0) Open fluorene.log in py.Aroma 5.
- (1) Check **Show Atom No.**.
- (2) Input 16 17 in the text field in **D** and click **Add Knot**.

- (3) Input 14 5 19 6 12 in the text field in **D** and click **Add Knot**.
- (4) Input 2 3 in the text bar in **D** and click **Add Knot**.
- (5) Change the value of **Interval** to 0.2.
- (6) Click **Save Input File**.
- (7) (option) Click **Save .png**. Finished!

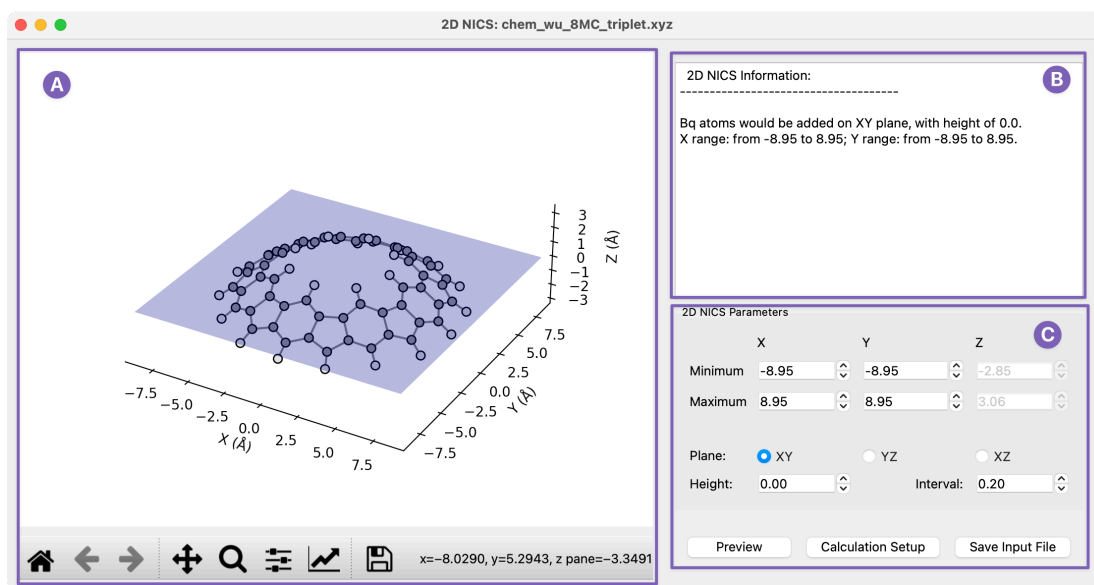


4.6 Create Input File for 2D NICS

4.6.1 Basic Information

The 2D NICS (ICSS) function can be accessed by clicking the **2D NICS** button after loading a file.

A provides a preview of the molecular structure and the region for NICS probes. Information about the probe region is summarized in **B**. In **C**, users can adjust the probe region, modify the height, and set the interval for ghost atoms.



If the 2D NICS parameters changed, the graph in **A** will not be updated unless the user clicks the **Preview** button in **C**.

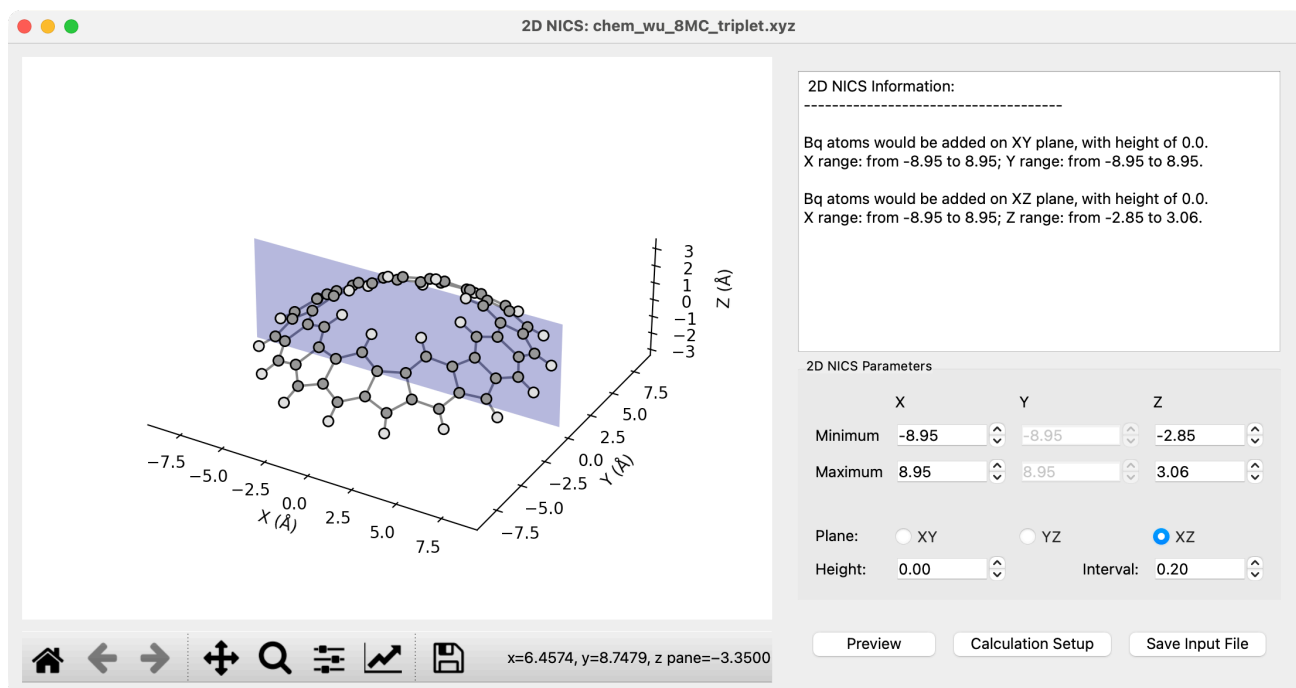
4.6.2 Example: AWA Molecule

Follows is a step-by-step tutorial for how to use 2D NICS module in py.Aroma 5. The molecule in this section is taken from **8MC** in *Chem*, **2018**, *4*, 1586 by Wu *et al.*

The molecule is placed parallel to XY plane, we want to put the NICS probes on the XY plane (target I) and XZ plane (target II), i.e., height is 0, to reproduce the Figure 3b in the original literature. The interval of 0.2 Å (default value in py.Aroma 5) would be used.

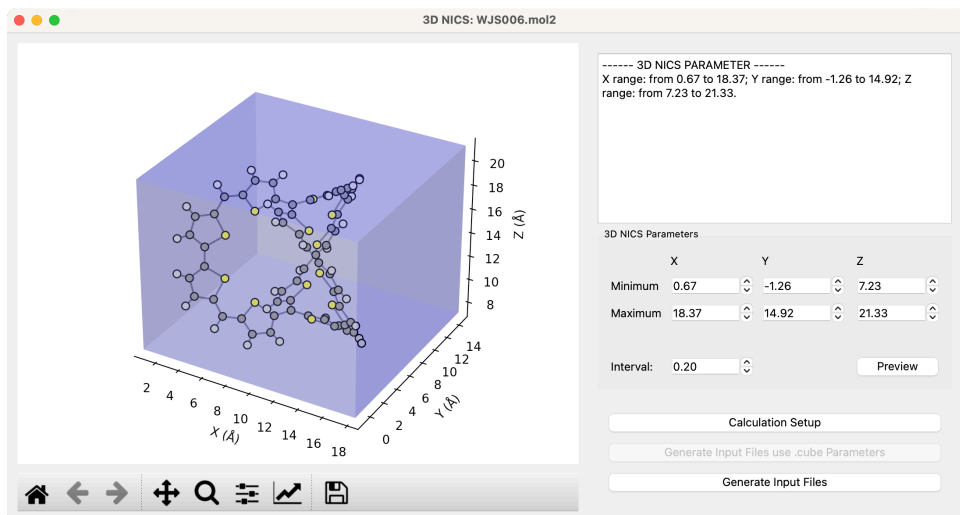
< Procedure >

- (0) Open `chem_wu_8MC_triplet.xyz` in py.Aroma 5.
- (1) Click **Save Input File**, .gjf file for the target I was saved, rename it with "_XY" suffix.
- (2) Choose **XZ** in **Plane** option, and the **Y** parameters would be disabled, and **Z** parameters would be enabled.
- (3) Click **Preview** to check the probe region.
- (4) Click **Save Input File**, .gjf file for the target II was saved, rename it with "_XZ" suffix.

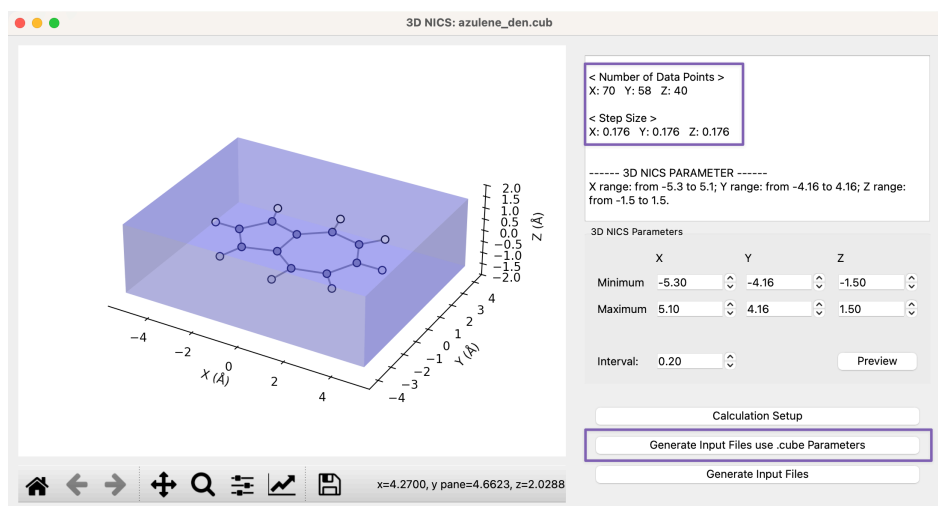


4.7 Create Input File for 3D NICS

Click **3D NICS** after loading a file. Usage mirrors the 2D NICS module (Section 4.6). Define the probe region by setting the X, Y, and Z limits and the probe spacing, then click **Save Input File**. Details are omitted here. An example, **c-T12⁴⁺** from *Nat. Chem.*, **2020**, *12*, 242 is shown below.



When opening a .cube (.cub) file in py.Aroma 5, you can generate 3D NICS input files that use the cube file's grid parameters. In the 3D NICS window, the grid parameters from the .cube file are displayed and the **Generate Input Files use .cube Parameters** button becomes available. Click this button to create the inputs; a parameter file named `_pm.txt` is saved in the same directory as the input files. This `_pm.txt` file is required when processing the corresponding output files in py.Aroma 5.

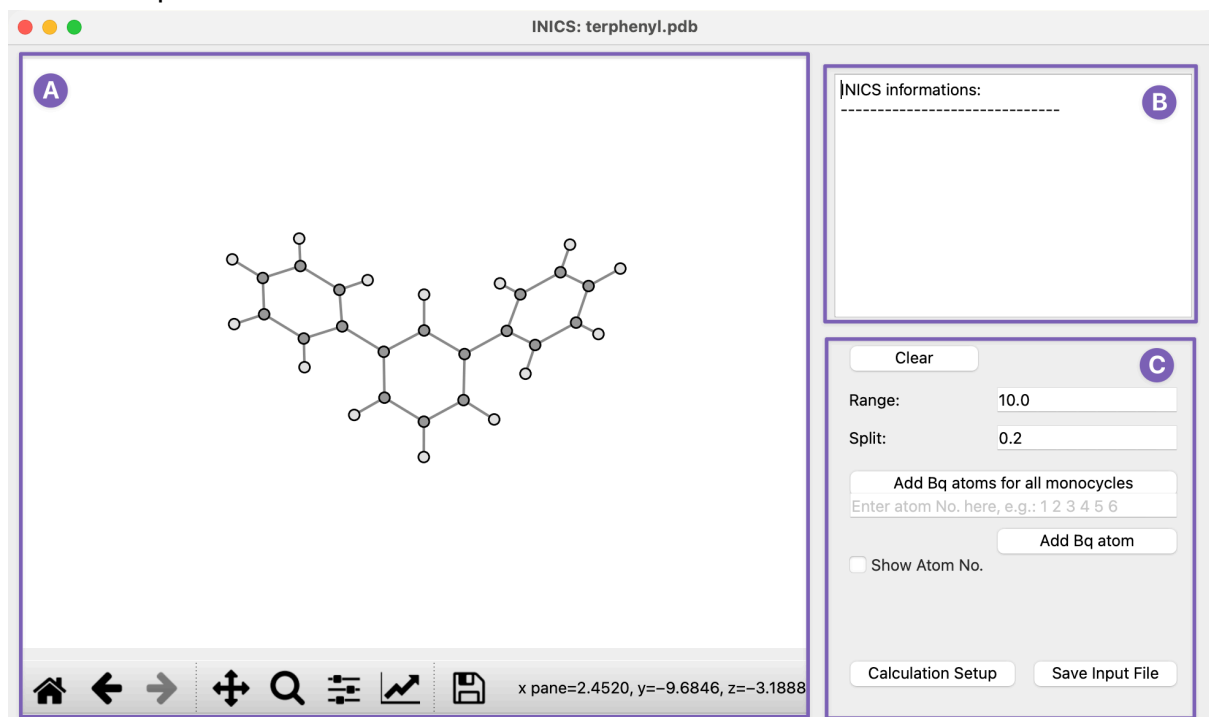


From Developer: *Gaussian* might have a limit on the number of ghost atoms, and it is somehow dependent of the version of *Gaussian* and the computer. To make *Gaussian* run normally, I set a limit to py.Aroma 5 on the number of ghost atoms to 7000 in a single .gjf file. Thus, especially for 3D NICS (ICSS), several files might be generated depends on the probe region, naming start from "`_0001.gjf`".

4.8 Create Input File for INICS

4.8.1 Basic Information

Click **INICS** after opening a file. The INICS window has three areas: **A** displays the molecular geometry, **C** lets users set INICS parameters and add ghost atoms, and **B** summarizes the defined INICS probes.



Specify the **range** and **split** (step size) in **C** to place ghost atoms. The default 10.0/0.2 adds probes from $-10 \text{ \AA}$ to $+10 \text{ \AA}$ along the ring normal at $0.2 \text{ \AA}$ spacing, which is sufficient for most systems. If multiple monocycles are present, click **Add Bq atoms for all monocycles** to place probes for every ring. To target a specific ring, enter its atom indices in the text field and click **Add Bq atom**, **A** updates to show the new probes.

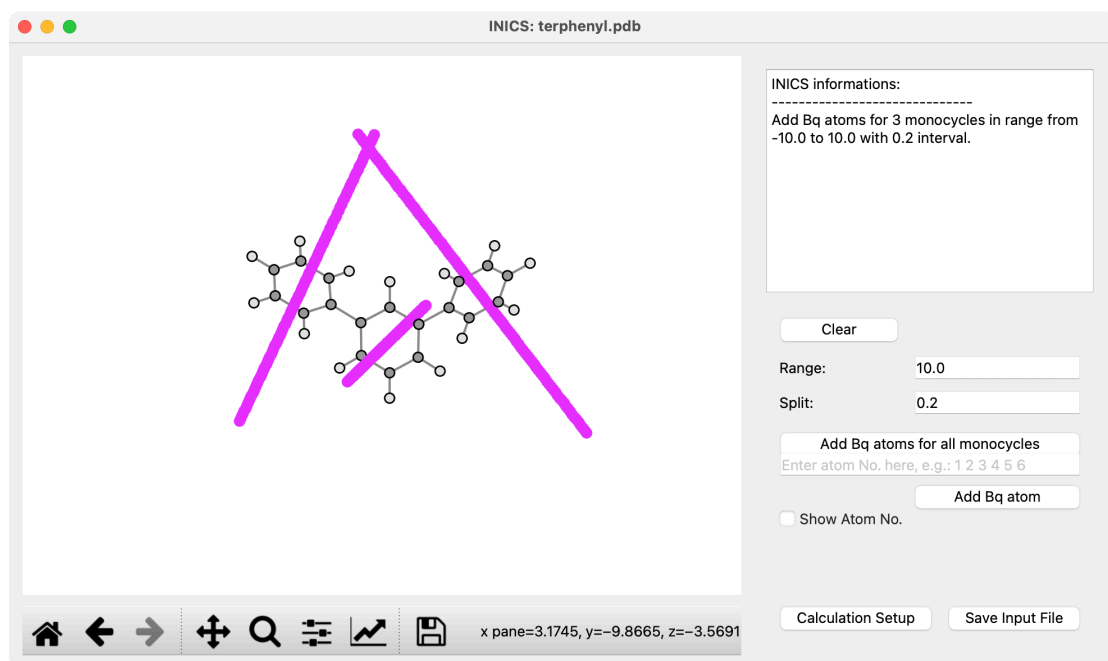
Note: py.Aroma 5 can generate INICS probes for arbitrary user-defined cycles, but outputs from such cycles cannot be analyzed in this version.

4.8.2 Example: Terphenyl

Follows is a step-by-step tutorial for how to use INICS module in py.Aroma 5. The target molecule is terphenyl, consist of three phenyl rings and two of terminal rings are identical in chemical environment. Our target is adding ghost atom from $(-10, +10)$, split = 0.2, using the default setting.

< Procedure >

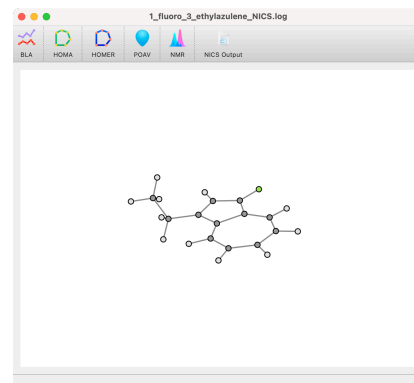
- (0) Open terphenyl.pdb in py.Aroma 5.
- (1) Click **Add Bq atoms for all monocycles**.
- (2) Click **Save Input File**.



4.9 Analyze Output Files for NICS Calculation

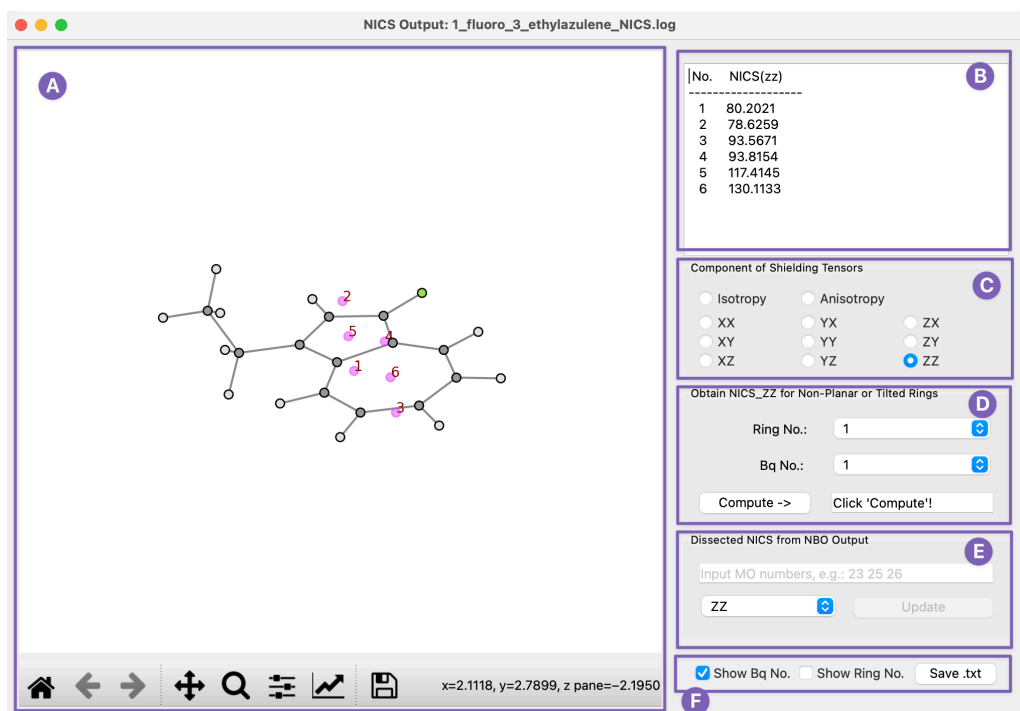
4.9.1 Basic Information

py.Aroma 5 automatically extracts shielding tensors from *Gaussian* output files. When an output includes ghost atoms, the main window shows the geometry with ghost atoms hidden. From this window users can open the BLA, HOMA, and POAV modules, and analyze NICS results by clicking **NICS Output**.



4.9.2 Single Point NICS

For NICS outputs, **A** shows the geometry with ghost atoms and **B** lists the extracted shielding tensors. To hide ghost-atom indices, uncheck **Show Bq No.** in **F**. The ZZ component is displayed in **B** by default, use the component radio buttons in **C** to switch components and **B** updates automatically. Export the data with **Save .txt** in **F**. **D** provides a pseudo-NICS_{ZZ} (i.e., NICS_⊥) calculation for nonplanar or tilted rings using the fitted ring normal. If an NBO dissected-NICS output is loaded, the **Dissected NICS from NBO Output** panel in **E** is enabled, enter molecular-orbital numbers to compute dissected quantities (e.g., NICS_π, NICS_σ).



NICS Output: 1_fluoro_3_ethylazulene_NICS.log

A

B

No.	NICS(zz)
1	80.2021
2	78.6259
3	93.5671
4	93.8154
5	117.4145
6	130.1133

C

Component of Shielding Tensors

☐ Isotropy ☐ Anisotropy
☐ XX ☐ YX ☐ ZX
☐ XY ☐ YY ☐ ZY
☐ XZ ☐ YZ ☒ ZZ

D

Obtain NICS_ZZ for Non-Planar or Tilted Rings

Ring No.: 1

Bq No.: 1

Compute -> Click 'Compute!'

E

Dissected NICS from NBO Output

Input MO numbers, e.g.: 23 25 26

ZZ Update

F

☒ Show Bq No. ☐ Show Ring No. Save .txt

x=2.1118, y=2.7899, z plane=-2.1950

For disordered ring or rings those are not parallel to Cartesian plane, py.Aroma 5 provides a useful function for compute pseudo-NICS_{ZZ} based on shielding tensors. Follows is an example for a terphenyl molecule, the middle phenyl ring (No. 2) is placed on the XY plane, whose NICS_{ZZ} will be the reference. The left phenyl ring (No. 1) has a dihedral angle with XY plane of 30°, and right phenyl ring (No. 3) is perpendicular to XY plane.

The $\text{NICS}(1)_{\text{ZZ}}$ of ring No. 2 is -26.0803 , indicating its high aromaticity. On the other hand, the $\text{NICS}(1)_{\text{ZZ}}$ from output file are -19.7001 and -0.7057 for ring No.1 and No. 3, respectively. Since the ring No. 1 and No. 3 are not placed on (or parallel) to XY plane, the ZZ shielding tensors are no longer reliable.

Compute the shielding tensor on the direction that perpendicular to the ring plane, i.e., $\text{NICS}_{\perp}$ in py.Aroma 5 is easy. Users just need to select the **ring No.** and **Bq No.** in **Obtain NICS_ZZ for Non-Planar or Tilted Rings** section, then click **Compute ->**. The pseudo- NICS_{ZZ} ($\text{NICS}_{\perp}$) value will be computed and displayed in the text field.

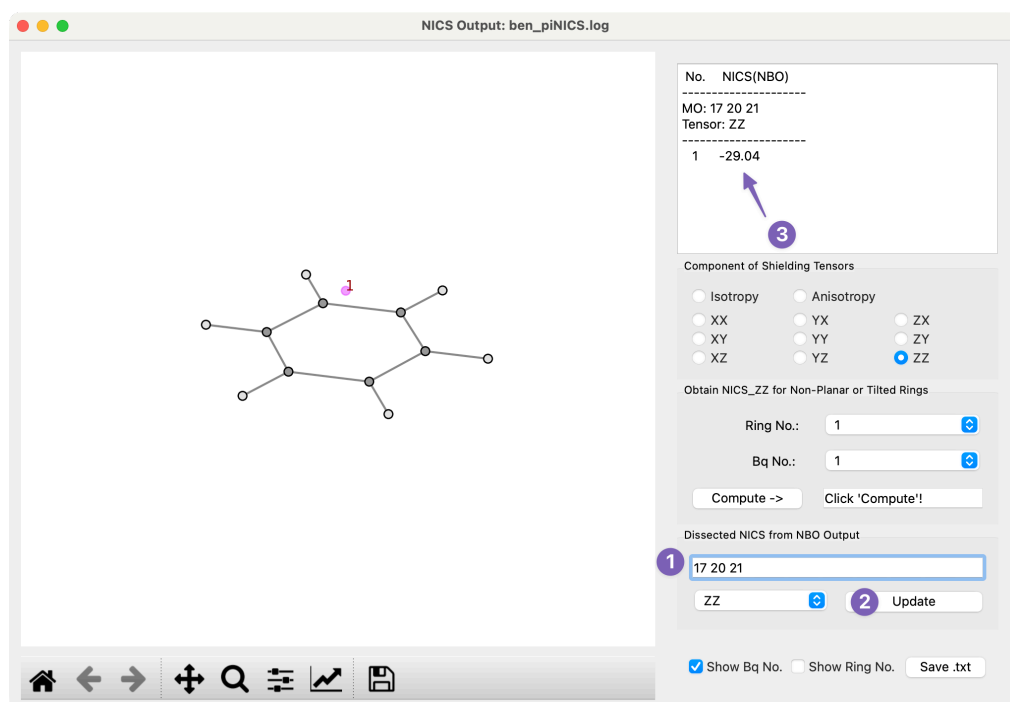
For example, the $\text{NICS}(1)_{\perp}$ of ring No. 1 and No. 2 were calculated to -26.3180 and -28.6472 , respectively, much close to $\text{NICS}(1)_{\text{ZZ}}$ of ring No. 2. Since ring No. 3 is nearly parallel to XZ plane, the $\text{NICS}(1)_{\text{YY}} = -26.6393$ could be treated as $\text{NICS}(1)_{\perp}$.

The screenshot shows the 'NICS Output: terphenyl_zz_NICS.log' window. On the left is a chemical structure of terphenyl with three rings numbered 1, 2, and 3. On the right is a table of NICS values:

No.	NICS(zz)
1	-19.7001
2	-19.6856
3	-26.0803
4	-26.0811
5	-0.7042
6	-0.7057
7	-8.6392
8	-10.6686
9	-6.0205

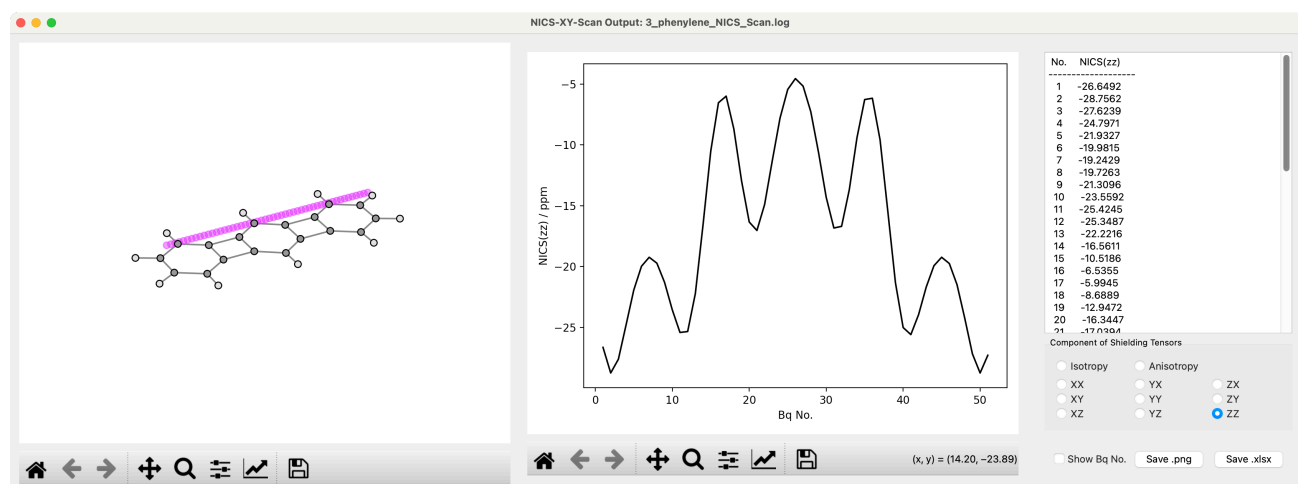
Below the table are sections for 'Component of Shielding Tensors' (Isotropy, Anisotropy with XX, XY, XZ, YX, YY, YZ, ZX, ZY, ZZ options), 'Obtain NICS_ZZ for Non-Planar or Tilted Rings' (Ring No.: 1, Bq No.: 1, Compute -> -26.3180), and 'Dissected NICS from NBO Output' (Input MO numbers, e.g.: 23 25 26, ZZ, Update). At the bottom right, there are checkboxes for 'Show Bq No.' and 'Show Ring No.' and a 'Save .txt' button.

For NICS calculations using the NBO additional keyword `$NBO NCS <XYZ MO> $END`, Gaussian provides per-MO contributions to the shielding tensor. In **E**, enter the MO numbers of interest (e.g., the π -MO set) and click **Update**, py.Aroma 5 sums those contributions and reports dissected indices such as NICS_{π} and NICS_{σ} for the selected tensor component (ZZ or isotropy). For example, following is a dissected NICS calculation for benzene, whose MO No. 17, No. 20 and No. 21 are assigned as π orbitals and the rest are σ orbitals. By inputting 17 20 21 in the text field in **E** and click **Update**, the $\text{NICS}(1)_{\pi\text{ZZ}}$ was calculated to be -29.04 .



4.9.3 NICS-XY-Scan

The main function of the NICS-XY-scan output is almost the same as NICS. The atom indices of ghost atoms are hidden by default for clarity, and the NICS-XY-scan plot is displayed in the middle part. When the component of the shielding tensor is changed, the NICS-XY-scan plot is also automatically updated. Users can save the plot by clicking the **Save .png** button and save the data to a .xlsx file by clicking the **Save .xlsx** button.

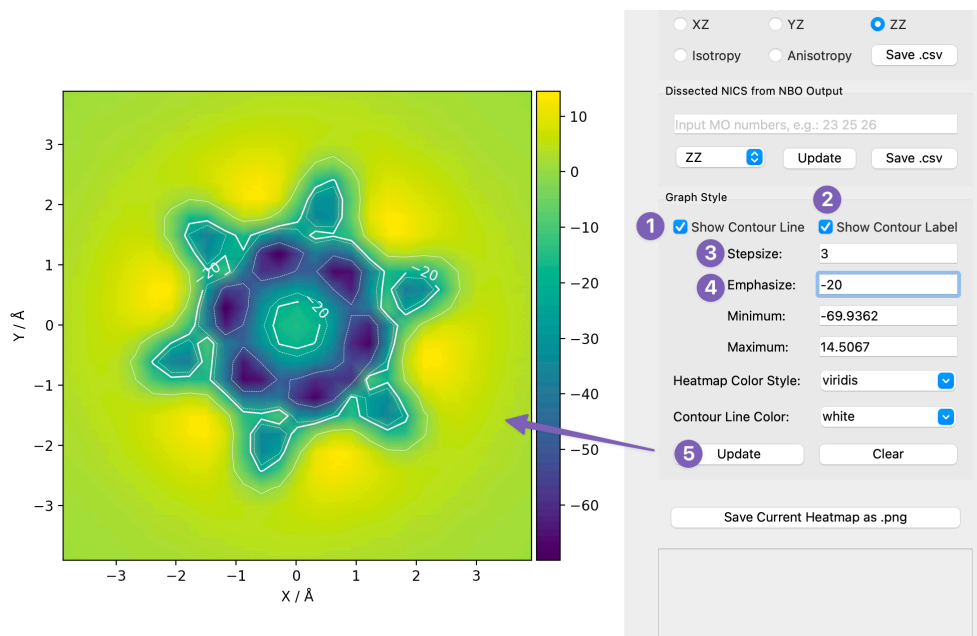
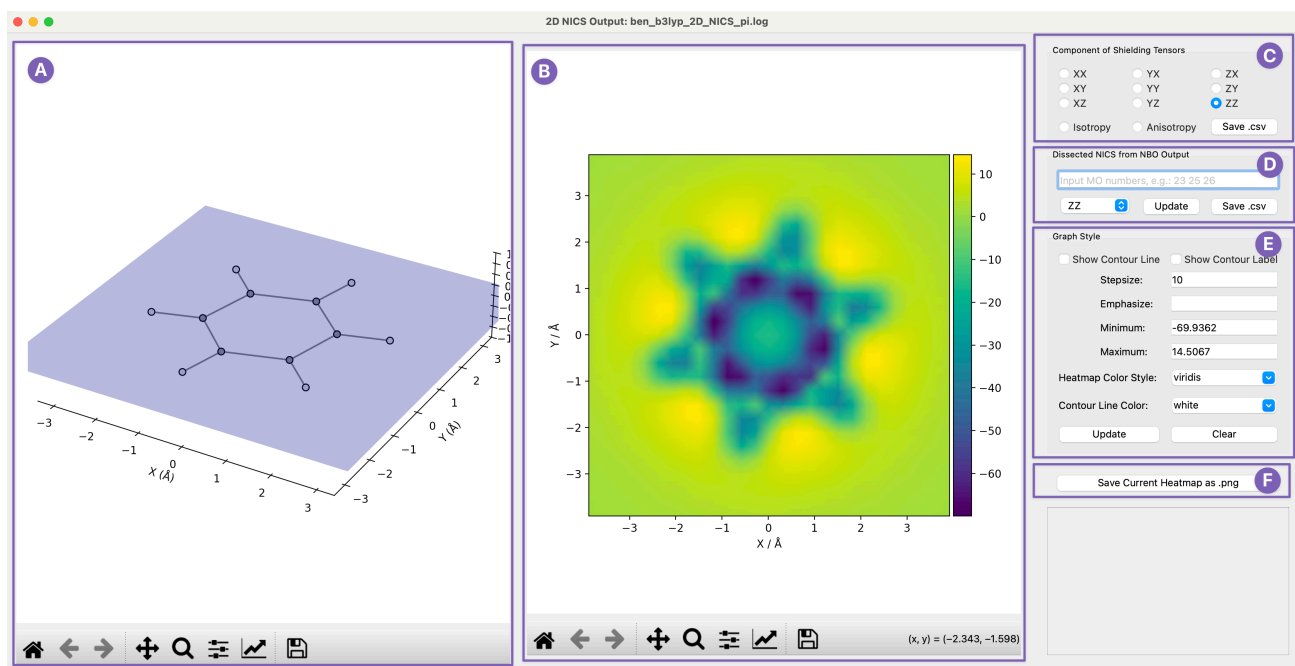


4.9.4 2D NICS/ICSS

In the 2D NICS/ICSS output window, **A** shows a preview of the molecule and the probe region (pale purple). **B** displays the 2D NICS/ICSS heatmap. Select the shielding-tensor component in **C**, the heatmap updates automatically. Use **Save Current Heatmap as .png** (in **F**) to export the

image and **Save .csv** to export the data (in **C** or **D**). Dissected NICS via NBO program is also supported for 2D maps.

In **E**, users can customize the heatmap. Check **Show Contour Line** to draw contours and set the interval in **Stepsize**. To highlight a specific contour, enter the value in **Emphasize** and click **Update**. Enable **Show Contour Label** to place a label on the emphasized contour. Users can also set the heatmap **minimum** and **maximum** values in **E**.



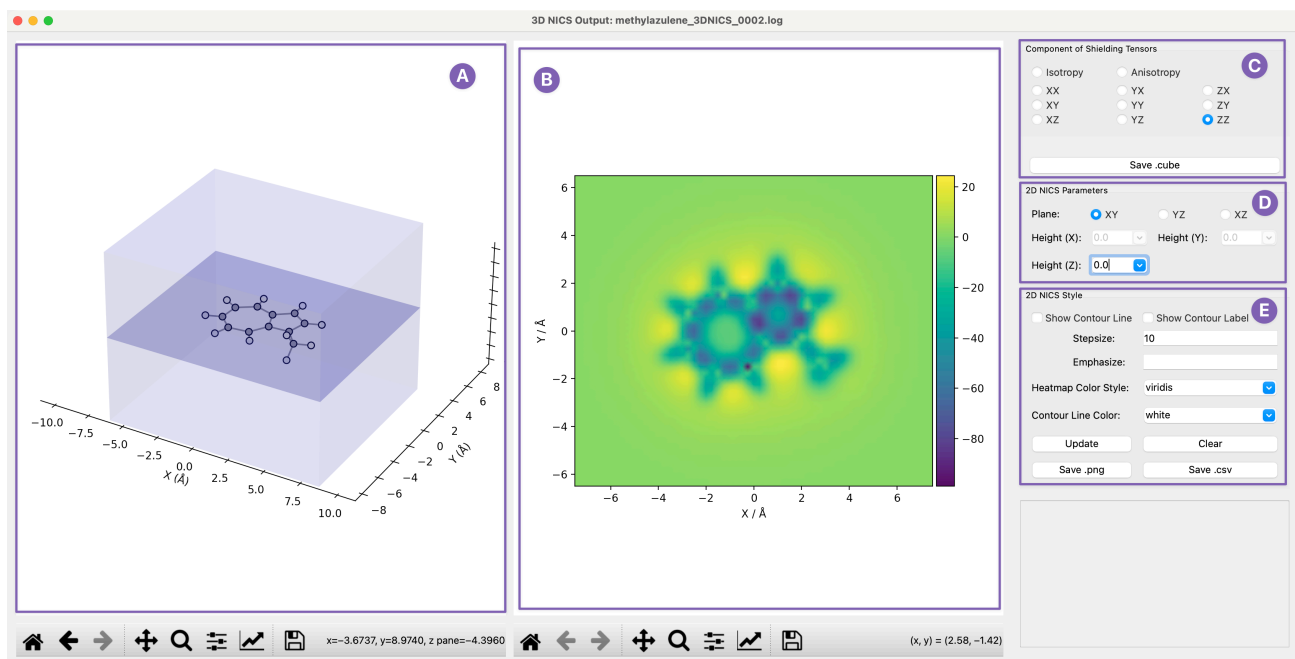
The heatmap function in py.Aroma 5 is powered by Matplotlib, so the color definition follows the rules of Matplotlib. Users can change the color style of the heatmap and the color of the contour line from the list in **E**. Colors not listed are also supported, as long as it follows the rule of Matplotlib,

and users can simply input the name of the color and click **Update**. For more information about color definition in Matplotlib, please refer to the Appendix I for colormap and Appendix II for single color specifying.

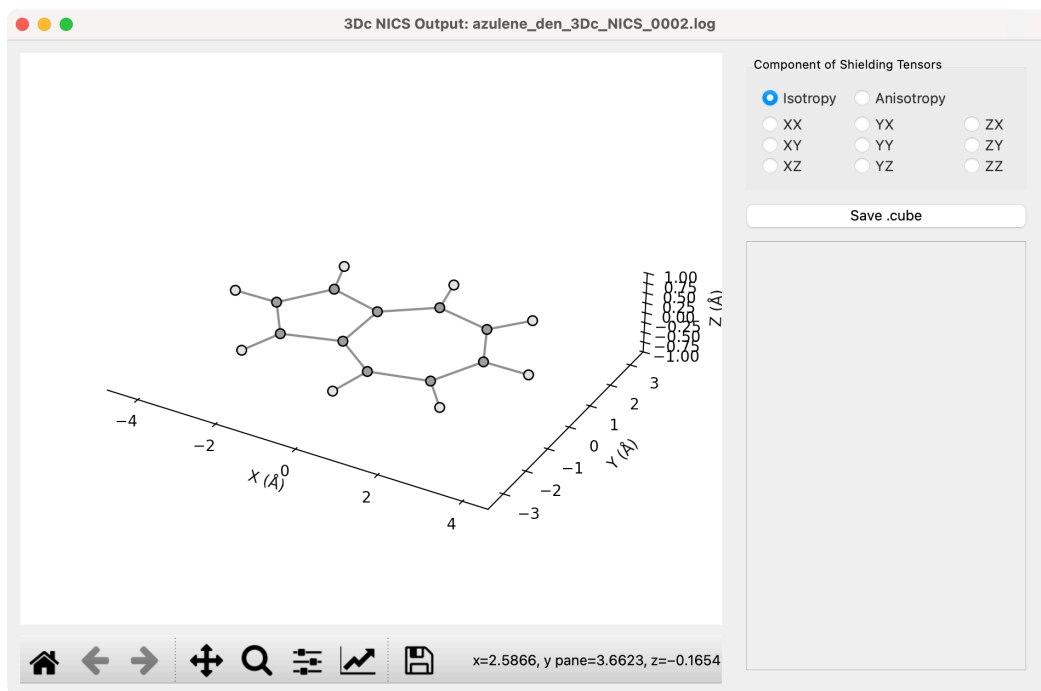
Note: For dissected NICS, set the heatmap options in **E**, then enter the MO numbers in **D** and click **Update** in **D**. Do not use **Update** in **E**, it changes styling only and does not recalculate the NICS data.

4.9.5 3D NICS/ICSS

In the 3D NICS/ICSS output window, **A** shows the molecule and the probe region. A 2D slice view is included in this module, so a heatmap on the XY plane is displayed by default in **B**. Use radio buttons in **D** to choose the **slice plane** (XY, YZ, XZ) and **height**, and use **E** to customize the plot style, matching the options in the 2D NICS/ICSS module. In **C**, select the shielding-tensor component to analyze; the view updates automatically, and you can export the data as a *Gaussian* cube file (.cube/.cub).

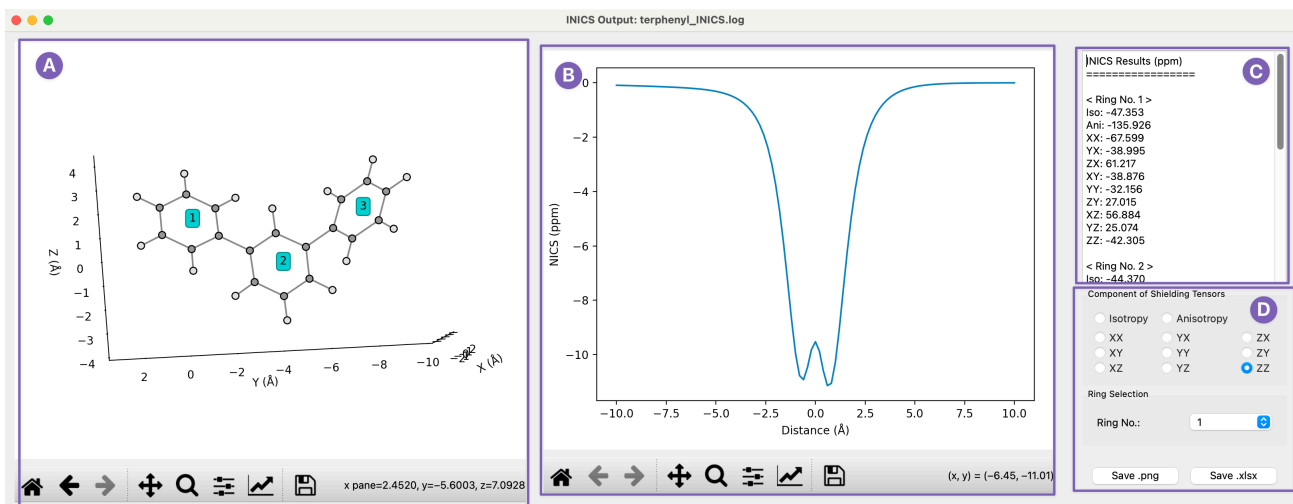


When inputs are generated from a .cube file (see Section 4.7), only the **Save tensors to .cube export is available**. Please note: the parameter file xx_3Dc_NICS_pm.txt created alongside the inputs, which containing the grid parameters and file header, is required to process the corresponding output files and must remain in the same directory.



4.9.6 INICS

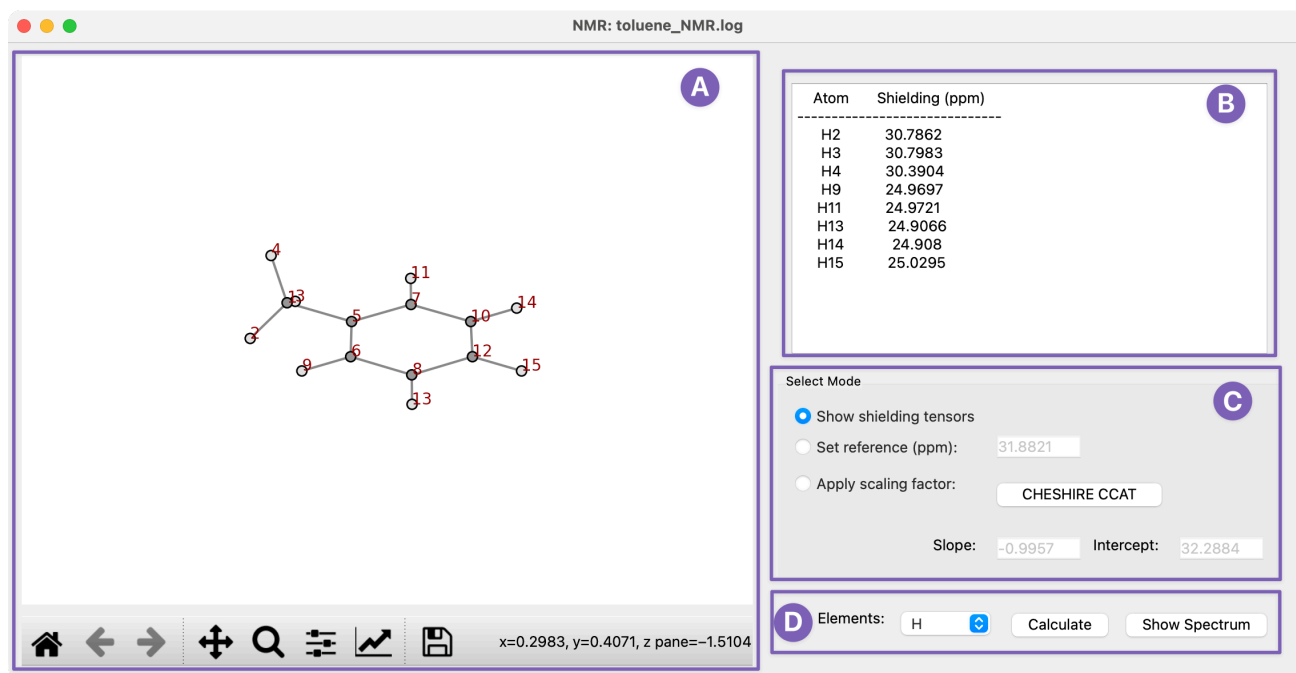
In the INICS output window, **A** shows the molecule with ring indices. **B** plots NICS vs. distance using NICS_{zz} for Ring 1 by default. Use **D** to select the tensor component and ring number, the plot updates accordingly. **C** lists the computed INICS values for all rings, for each tensor component. Use the buttons in **D** to save the plot as a .png and export all shielding tensors as an .xlsx file.



4.10 NMR Plotting

The py.NMR module would be active when reading a *Gaussian* output file of NMR job. The output files for NICS analyses are also possible to active py.NMR module.

4.10.1 Main Function



Main window of py.NMR module is shown as above. Just like other modules, the geometry (without ghost atoms) will be displayed in (A). The isotropic shielding tensors will be displayed in (B). Users can select the element of interest in the molecule from the "Element" list in (D).

py.NMR module provides three modes to process shielding tensors, users can choose from (C). **Show shielding tensors** will do nothing on the isotropic shielding tensors. **Set reference** allows users to input a reference value to compute chemical shift, the default value is shielding tensor of protons in trimethylsilane (TMS) calculated at B3LYP/6-311+G(2d,p) level of theory. **Apply scaling factor** allows users to scale the chemical shift based on the empirical scaling factors. The scaling factor could be found in CHESHIRE CCAT³ website, and users can easily access the website by click **CHESHIRE CCAT** button.

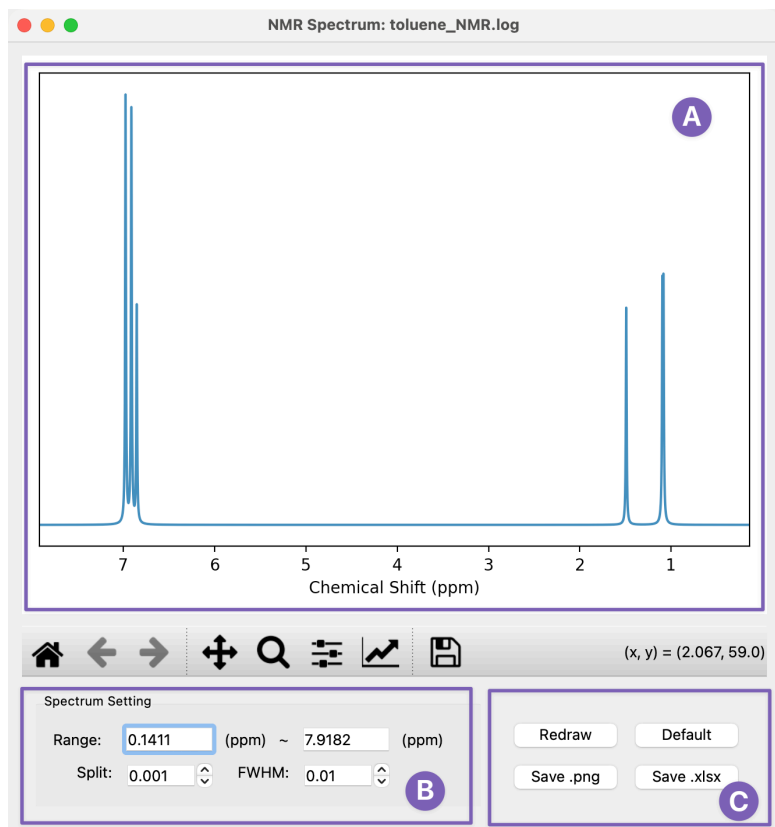
Users need to click **Calculate** in (D) to update the shielding tensors/chemical shift if the mode or element changed, the new results will be updated in (B). And users can also check the NMR spectrum by clicking **Show Spectrum** button.

4.10.2 NMR Spectrum

In NMR spectrum window, the NMR spectrum will be plotted in (A). Users can modify the plotting parameters like **plot range**, **split** and **FWHM** in (B). A smaller split value will give a smoother

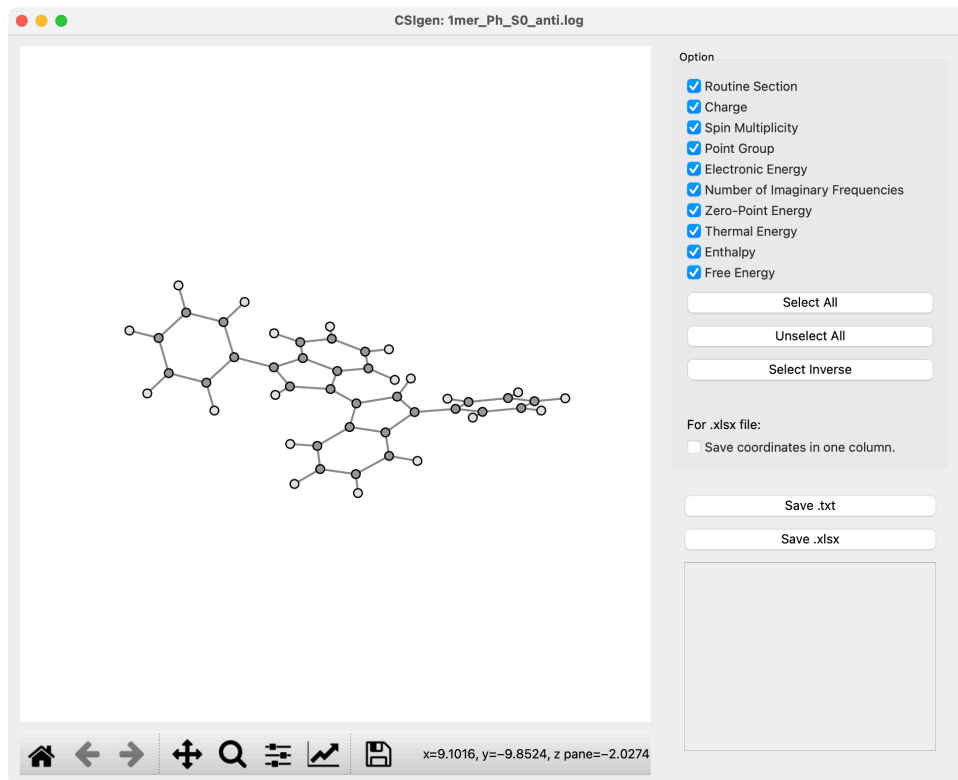
³ <http://cheshirenmr.info/ScalingFactors.htm>

spectrum, and a smaller FWHM value will give sharper peaks (See Appendix IV). If parameters changed, users are required to click **Redraw** button to update the spectrum, and **Default** button allows user use default parameters. The NMR spectrum can be saved as .png and .xlsx via the buttons in **C**.



4.11 Generate Computational Supporting Material

The CSIgen module would be active when reading a *Gaussian* output file without ghost atoms to py.Aroma 5. The main window of CSIgen module is show below.

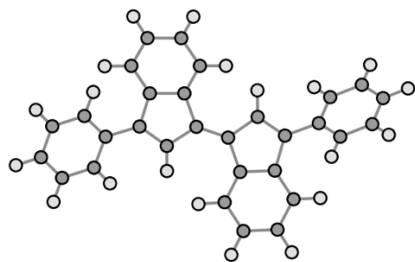


The geometry of the molecule is displayed in left area. In the check list at right part, users can choose the information needed for the supporting information. By default, all information is selected. The options **Number of Imaginary Frequencies**, **Zero-Point Energy**, **Thermal Energy**, **Enthalpy** and **Free Energy** are only available when a `freq` job is detected.

Users can choose to save the supporting information as a `.txt` or `.xlsx` file. When saving as a `.xlsx` file, the Cartesian coordinates are saved in two columns by default, but this can be turned off by checking the **Save coordinates in one column** option.

The follows page provides an example using the CSIgen module in py.Aroma 5 to generate supporting information. The geometry in the upper left was added later.

Hint: When saving the coordinates in two columns, the width is just to fit the **Moderate** margins setting in *Microsoft Word*.

1mer_Ph_S0_anti

#p opt freq rwb97xd/6-31g(d,p)

Charge = 0, Multiplicity = 1, Point group = C1

Number of imaginary frequencies = 0

Electronic Energy = -1154.87028937 Hartree

Sum of electronic and zero-point Energies = -1154.463745 Hartree

Sum of electronic and thermal Energies = -1154.441758 Hartree

Sum of electronic and thermal Enthalpies = -1154.440814 Hartree

Sum of electronic and thermal Free Energies = -1154.516642 Hartree

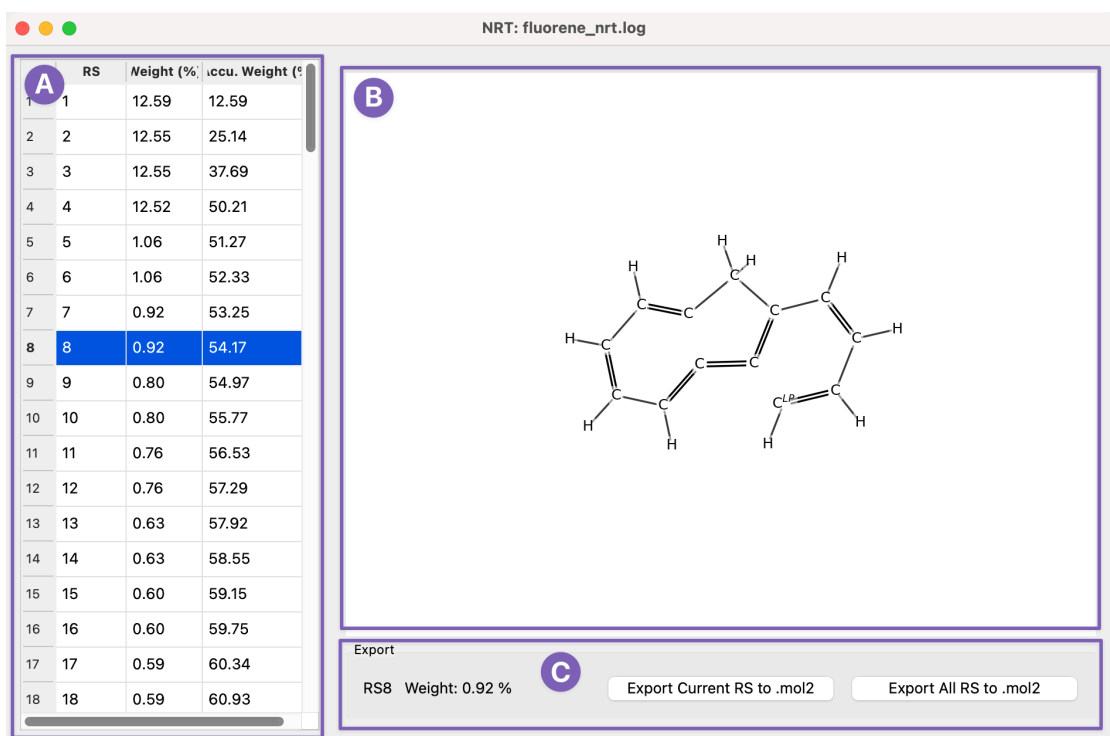
Cartesian Coordinates				Cartesian Coordinates			
Atoms	X	Y	Z	Atoms	X	Y	Z
H	4.435677	-2.606388	-0.231397	C	3.356437	-2.664594	-0.328848
C	0.569521	-2.799826	-0.699768	C	2.571309	-1.527073	-0.212788
C	2.735491	-3.884182	-0.603806	C	1.358363	-3.948847	-0.791078
C	1.171317	-1.589434	-0.381275	H	3.334927	-4.784470	-0.693673
H	0.890493	-4.899083	-1.027388	H	-0.491294	-2.867073	-0.908157
C	2.946798	-0.115707	-0.003726	C	1.825138	0.639775	-0.089644
H	1.789433	1.700971	0.109068	C	0.648524	-0.205519	-0.281946
H	-4.435780	2.606331	-0.231253	C	-3.356547	2.664605	-0.328741
C	-0.569663	2.800007	-0.699804	C	-2.571346	1.527130	-0.212737
C	-2.735690	3.884236	-0.603706	C	-1.358578	3.948982	-0.791055
C	-1.171361	1.589573	-0.381277	H	-3.335187	4.784487	-0.693533
H	-0.890777	4.899247	-1.027383	H	0.491130	2.867330	-0.908265
C	-2.946755	0.115735	-0.003721	C	-1.825064	-0.639688	-0.089728
H	-1.789301	-1.700902	0.108896	C	-0.648491	0.205678	-0.281947
C	-4.302591	-0.380861	0.272258	C	-6.864325	-1.376529	0.828160
C	-5.157951	0.298025	1.149401	C	-4.753603	-1.566015	-0.321422
C	-6.022475	-2.060311	-0.044873	C	-6.427119	-0.197450	1.425233
H	-4.812770	1.203896	1.637694	H	-4.104710	-2.086849	-1.019008
H	-6.358149	-2.977689	-0.518023	H	-7.074646	0.336285	2.113683
H	-7.856805	-1.760298	1.041443	C	4.302664	0.380807	0.272255
C	6.864458	1.376320	0.828162	C	5.158007	-0.298169	1.149344
C	4.753723	1.565971	-0.321369	C	6.022624	2.060190	-0.044816
C	6.427205	0.197228	1.425180	H	4.812791	-1.204049	1.637593
H	4.104843	2.086876	-1.018915	H	6.358333	2.977577	-0.517923
H	7.074719	-0.336576	2.113587	H	7.856959	1.760028	1.041449

4.12 Natural Resonance Theory (NRT) Analysis

The natural resonance theory (NRT) analysis is an option in the NBO program invoked with the additional keyword \$NBO NRT \$END. It enumerates Lewis resonance structures (RS) and assigns each a resonance weight.

When an output containing NRT data is opened in py.Aroma 5, the NRT module becomes available and provides quick visualization of the resonance structures:

- A RS List** — displays each RS with its weight (%) and cumulative weight (%). Click an entry to select it.
- B Resonance structure view** — shows the structure of the selected RS.
- C Export** — export the current RS or all RS entries as .mol2 files.



NRT: fluorene_nrt.log

RS	Weight (%)	cum. Weight (%)
1	12.59	12.59
2	12.55	25.14
3	12.55	37.69
4	12.52	50.21
5	1.06	51.27
6	1.06	52.33
7	0.92	53.25
8	0.92	54.17
9	0.80	54.97
10	0.80	55.77
11	0.76	56.53
12	0.76	57.29
13	0.63	57.92
14	0.63	58.55
15	0.60	59.15
16	0.60	59.75
17	0.59	60.34
18	0.59	60.93

Export

RS8 Weight: 0.92 %

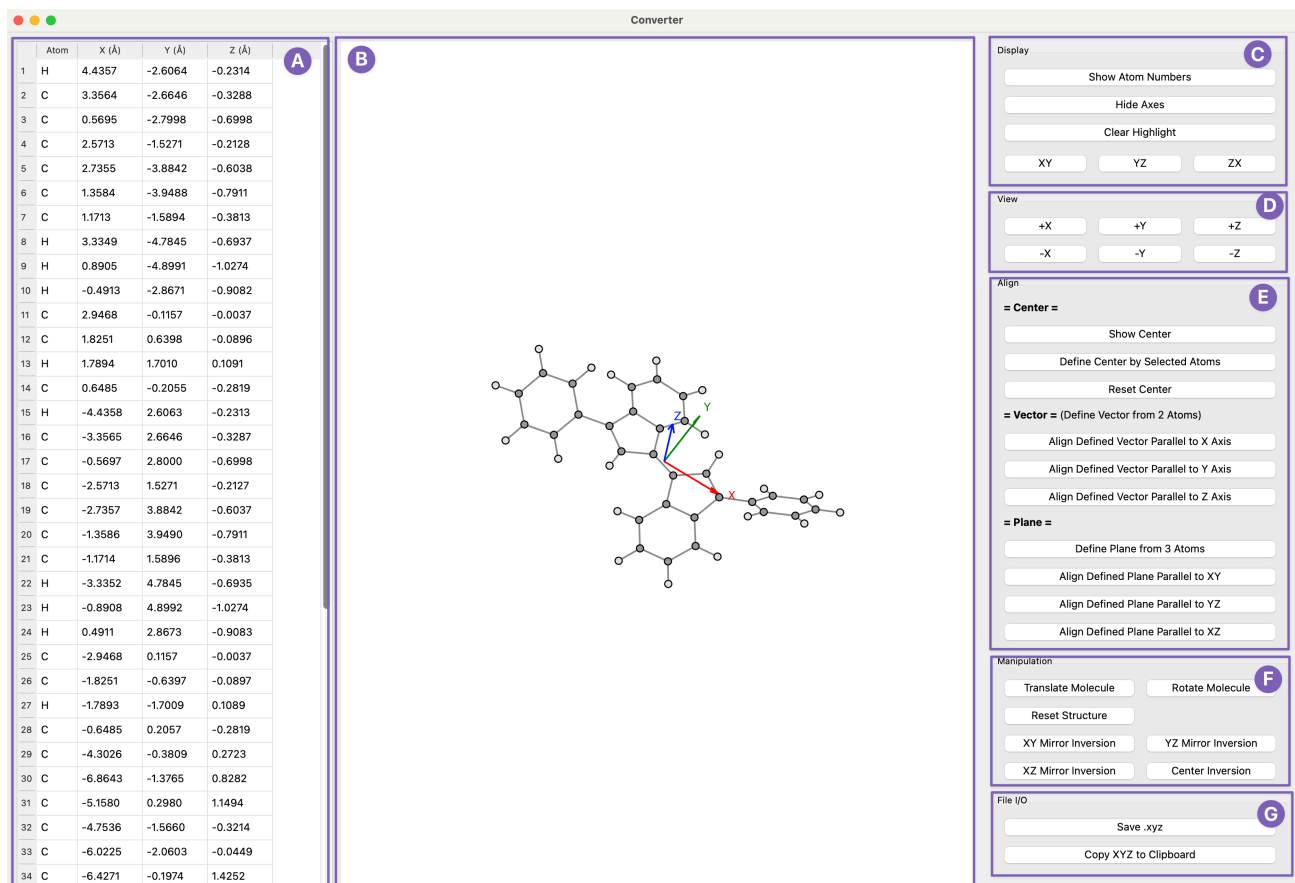
Export Current RS to .mol2

Export All RS to .mol2

4.13 Geometry Operation

py.Aroma 5 includes a new **Geometry Operations** module for editing molecular structures. The main window has:

- A Coordinates** — lists atomic Cartesian coordinates. Clicking an entry highlights the atom and enables relevant edit tools.
- B 3D view** — shows an interactive model of the molecule with Cartesian axes centered on the structure.



	Atom	X (Å)	Y (Å)	Z (Å)
1	H	4.4357	-2.6064	-0.2314
2	C	3.3564	-2.6646	-0.3288
3	C	0.5695	-2.7998	-0.6998
4	C	2.5713	-1.5271	-0.2128
5	C	2.7355	-3.8842	-0.6038
6	C	1.3584	-3.9488	-0.7911
7	C	1.1713	-1.5894	-0.3813
8	H	3.3349	-4.7845	-0.6937
9	H	0.8905	-4.8991	-1.0274
10	H	-0.4913	-2.8671	-0.9082
11	C	2.9468	-0.1157	-0.0037
12	C	1.8251	0.6398	-0.0896
13	H	1.7894	1.7010	0.1091
14	C	0.6485	-0.2055	-0.2819
15	H	-4.4358	2.6063	-0.2313
16	C	-3.3565	2.6646	-0.3287
17	C	-0.5697	2.8000	-0.6998
18	C	-2.5713	1.5271	-0.2127
19	C	-2.7357	3.8842	-0.6037
20	C	-1.3586	3.9490	-0.7911
21	C	-1.1714	1.5896	-0.3813
22	H	-3.3352	4.7845	-0.6935
23	H	-0.8908	4.8992	-1.0274
24	H	0.4911	2.8673	-0.9083
25	C	-2.9468	0.1157	-0.0037
26	C	-1.8251	-0.6397	-0.0897
27	H	-1.7893	-1.7009	0.1089
28	C	-0.6485	0.2057	-0.2819
29	C	-4.3026	-0.3809	0.2723
30	C	-6.8643	-1.3765	0.8282
31	C	-5.1580	0.2980	1.1494
32	C	-4.7536	-1.5660	-0.3214
33	C	-6.0225	-2.0603	-0.0449
34	C	-6.4271	-0.1974	1.4252

Area **C** to **G** provides some useful functions to modify the geometry:

- C Display** — show or hide atom numbers, axes, highlighted atoms, and Cartesian planes.
- D View** — set the viewing direction along the X, Y, or Z axis.
- E Align** — set the molecular center; align a vector defined by two atoms parallel to a chosen axis; align a plane defined by three atoms parallel to a chosen Cartesian plane.
- F Manipulation** — translate, rotate, mirror the geometry, or reset to the original.
- G File I/O** — export the edited structure as an .xyz file or copy Cartesian coordinates to the clipboard.

5 License



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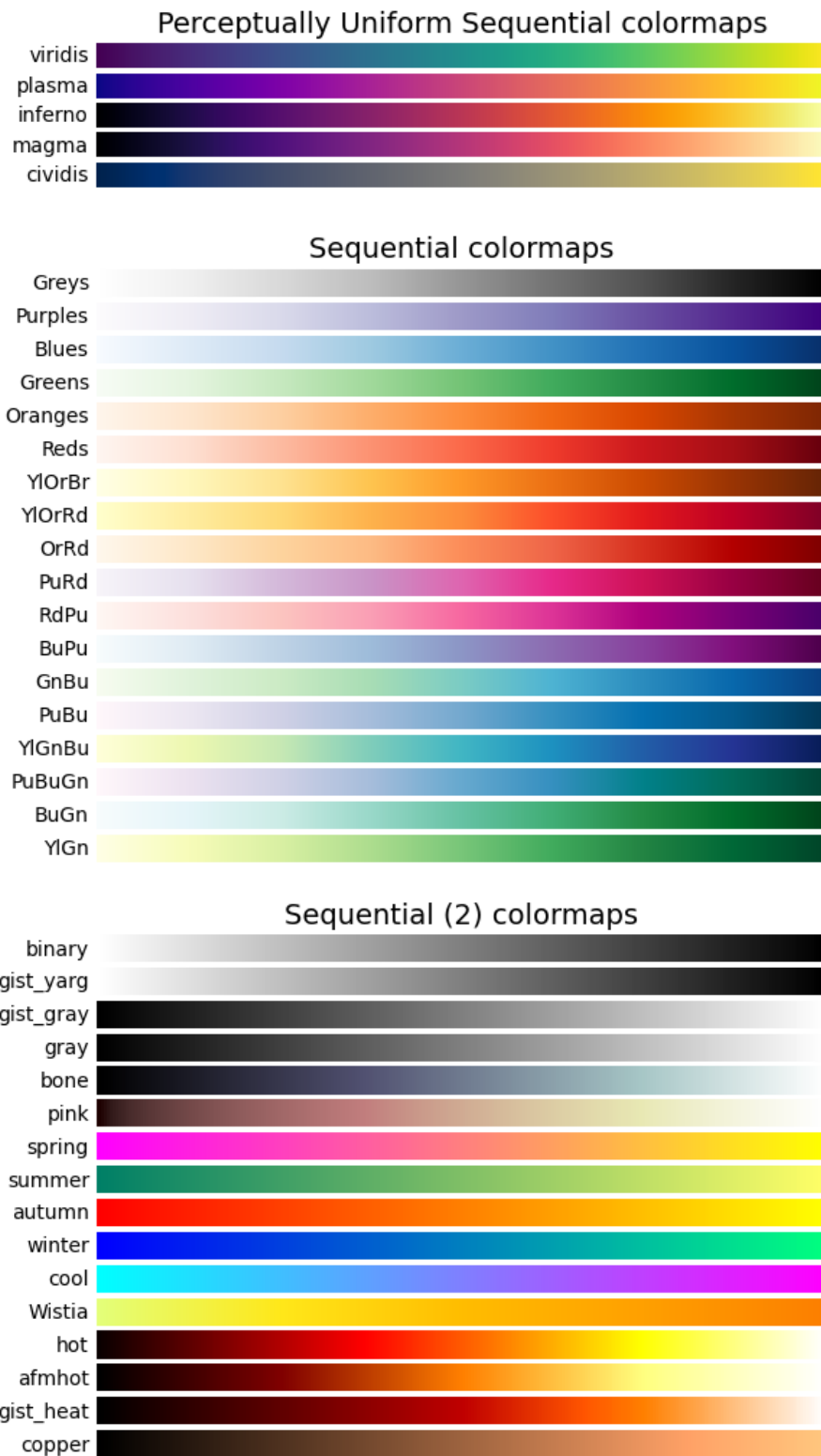
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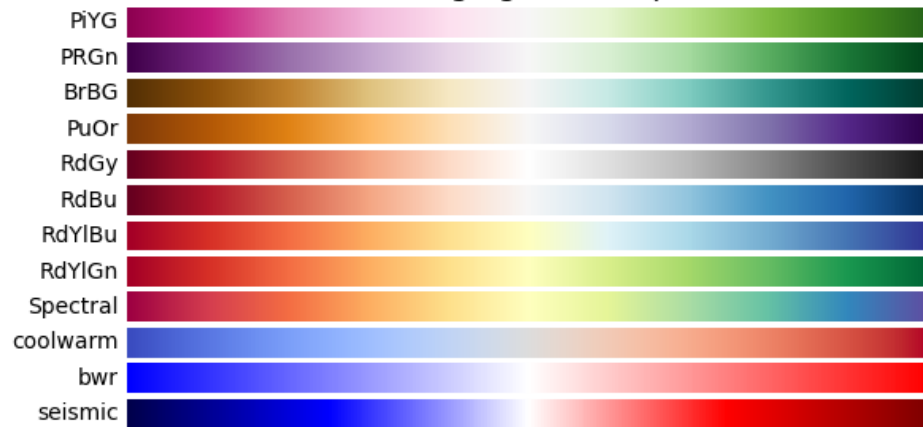
IMPORTANT: See **Appendix VII** for the full license contents.

Appendix

I Colormaps in Matplotlib



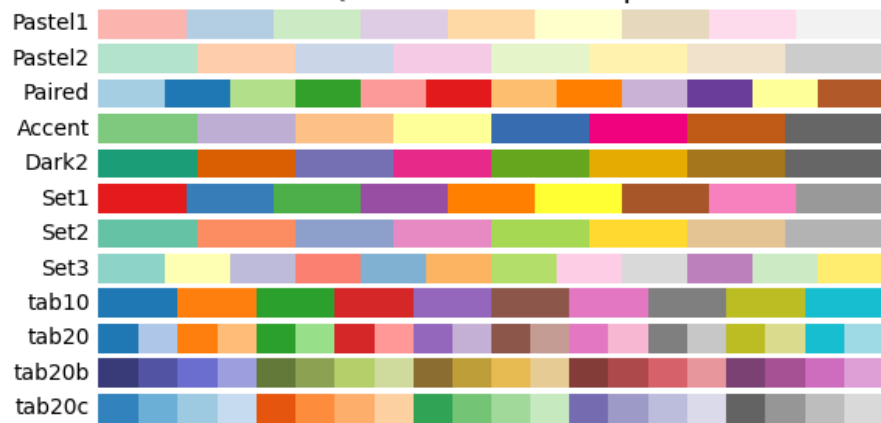
Diverging colormaps



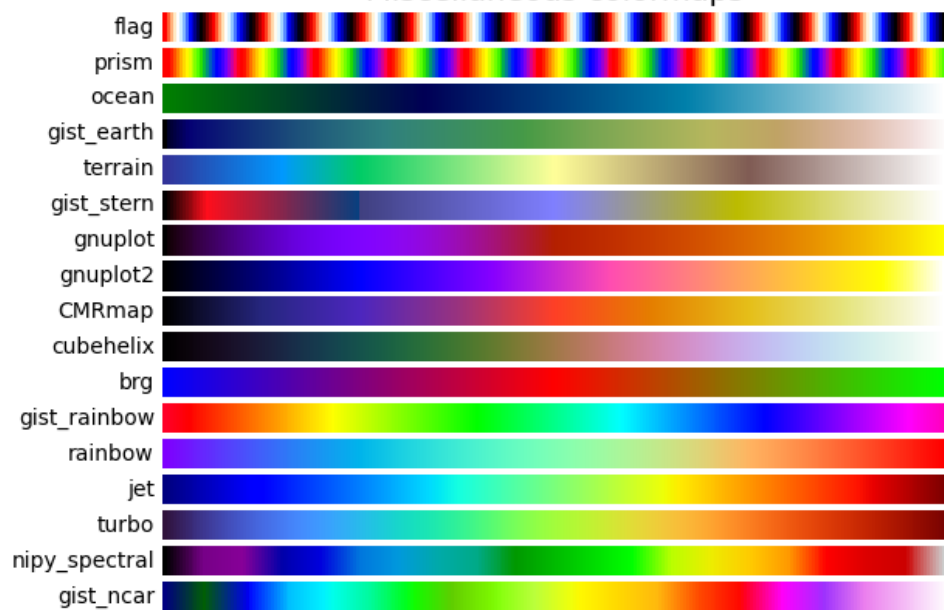
Cyclic colormaps



Qualitative colormaps



Miscellaneous colormaps



II Colors in Matplotlib

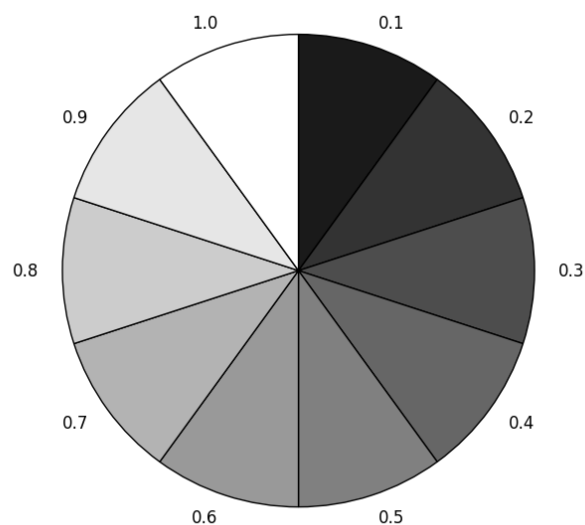
Follows contents are taken from <https://matplotlib.org/stable/tutorials/colors/colors.html>, Matplotlib recognizes the follows formats to specify a color.

Format	Example
RGB or RGBA (red, green, blue, alpha) tuple of float values in a closed interval [0, 1].	• (0.1, 0.2, 0.5) • (0.1, 0.2, 0.5, 0.3)
Case-insensitive hex RGB or RGBA string.	• '#0f0f0f' • '#0f0f0f80'
Case-insensitive RGB or RGBA string equivalent hex shorthand of duplicated characters.	• '#abc' as '#aabbcc' • '#fb1' as '#ffbb11'
String representation of float value in closed interval [0, 1] for grayscale values.	• '0' as black • '1' as white • '0.8' as light gray
Single character shorthand notation for some basic colors.	• 'b' as blue • 'g' as green • 'r' as red • 'c' as cyan • 'm' as magenta • 'y' as yellow • 'k' as black • 'w' as white
Case-insensitive X11/CSS4 color name with no spaces.	• 'aquamarine' • 'mediumseagreen'
Case-insensitive color name from xkcd color survey with 'xkcd:' prefix.	• 'xkcd:sky blue' • 'xkcd:eggshell'
Case-insensitive Tableau Colors from 'T10' categorical palette.	• 'tab:blue' • 'tab:orange' • 'tab:green' • 'tab:red' • 'tab:purple' • 'tab:brown' • 'tab:pink' • 'tab:gray' • 'tab:olive' • 'tab:cyan'
"CN" color spec where 'C' precedes a number acting as an index into the default property cycle.	• 'C0' • 'C1'

And here are some color names available in matplotlib.

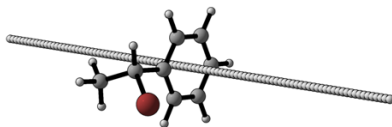
black	dimgray	dimgray	gray
grey	darkgrey	darkgray	silver
lightgray	lightgrey	gainsboro	whitesmoke
white	snow	rosybrown	lightcoral
indianred	brown	firebrick	maroon
darkred	red	mistyrose	salmon
tomato	darksalmon	coral	orangered
lightsalmon	sienna	seashell	chocolate
saddlebrown	sandybrown	peachpuff	peru
linen	bisque	darkorange	burlywood
antiquewhite	tan	navajowhite	blanchedalmond
papayawhip	moccasin	orange	wheat
oldlace	floralwhite	darkgoldenrod	goldenrod
cornsilk	gold	lemonchiffon	khaki
palegoldenrod	darkkhaki	ivory	beige
lightyellow	lightgoldenrodyellow	olive	yellow
olivedrab	yellowgreen	darkolivegreen	greenyellow
chartreuse	lawngreen	sage	lightsage
darksage	honeydew	darkseagreen	palegreen
lightgreen	forestgreen	limegreen	darkgreen
green	lime	seagreen	mediumseagreen
springgreen	mintcream	mediumspringgreen	mediumaquamarine
aquamarine	turquoise	lightseagreen	mediumturquoise
azure	lightcyan	paleturquoise	darkslategray
darkslategrey	teal	darkcyan	aqua
cyan	darkturquoise	cadetblue	powderblue
lightblue	deepskyblue	skyblue	lightskyblue
steelblue	aliceblue	dodgerblue	lightslategrey
lightslategray	slategray	slategrey	lightsteelblue
cornflowerblue	royalblue	ghostwhite	lavender
midnightblue	navy	darkblue	mediumblue
blue	slateblue	darkslateblue	mediumslateblue
mediumpurple	blueviolet	indigo	darkorchid
darkviolet	mediumorchid	thistle	plum
violet	purple	darkmagenta	fuchsia
magenta	orchid	mediumvioletred	deeppink
hotpink	lavenderblush	palevioletred	crimson
pink	lightpink		

About grayscale values:



III Accuracy of Integral NICS

Effects of range and interval on INICS index were investigated using (1-bromoethyl)benzene. As shown in follows, the default setting (Range: -10 to 10 Å, interval: 0.2 Å) provides well results.



Range	Interval / Å	INICS / ppm
-10 to 10 Å	0.1	-45.18
-10 to 10 Å*	0.2^*	-45.18
-10 to 10 Å	0.5	-45.18
-8 to 8 Å	0.2	-44.81
-5 to 5 Å	0.2	-43.29
-2 to 2 Å	0.2	-33.21

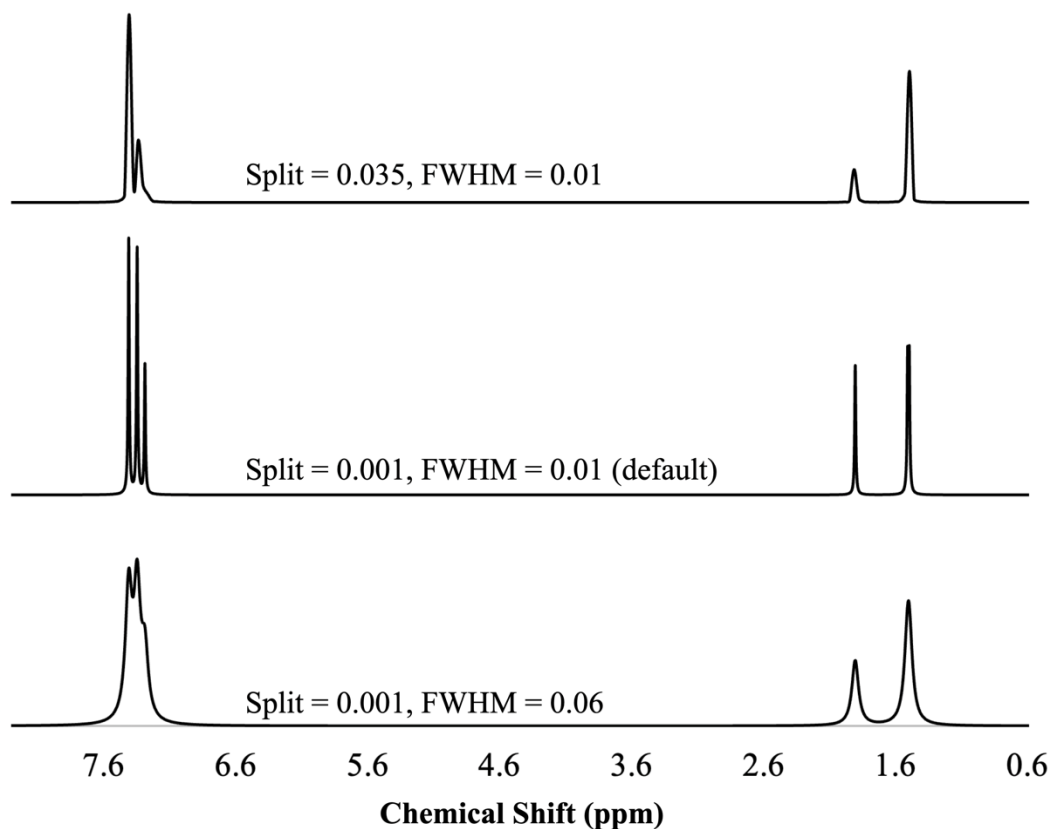
*Default setting in py.Aroma 5

IV Lorentzian Line Broadening in NMR Spectrum

NMR spectrum follows Lorentzian function for line broadening:

$$L(x) = \frac{\text{FWHM}}{2\pi} \frac{1}{(x - x_i)^2 + 0.25 \times \text{FWHM}^2}$$

where, FWHM indicates **Full Width at Half Maximum** and x_i indicates the chemical shift of target peak. Follows are examples of how **split** and **FWHM** affect the line shape in NMR spectrum.



V Adding Bond Parameters to HOMA and HOMER Analyses

Adding custom bond types to HOMA and HOMER modules in py.Aroma 5 requires manual editing of source code. The following example demonstrates adding O–Se bond support (set to $R_{\text{opt}} = 2.0$, $\alpha = 166.0$). Please replace the red text that match your needs.

1. Modify the configuration file

- (1) Open the config.ini file with text editor.
- (2) Locate the [homa] or [homer] section, depending on which component you're working with.
- (3) Add the following line to the section:

```
ose = 2.0,166.0
```

Steps 2 ~ 6 involve modifying the pyaroma_main.py file.

2. Add code for reading parameters form the configuration file

2.1 For HOMA

Insert following code after line 1386. Bonds those formed by the same elements only need to be added once.

e.g., O–Se bond:

```
'OSE':[float(config_file.get('homa', 'ose').split(',')[0]), \
        float(config_file.get('homa', 'ose').split(',')[1])],
'SEO':[float(config_file.get('homa', 'ose').split(',')[0]), \
        float(config_file.get('homa', 'ose').split(',')[1])]
```

e.g., O–O bond:

```
'OO':[float(config_file.get('homa', 'oo').split(',')[0]), \
        float(config_file.get('homa', 'oo').split(',')[1])]
```

2.2 For HOMER

Add similar code as 2.1 but replace "homa" with "homer" in the config_file.get() function calls.

3. Define global variables

Add global oseRopt, oseAlpha after line 9070 for HOMA; add global oseRoptHomer, oseAlphaHomer after line 9241 for HOMER.

Steps 4 ~ 6 are optional for displaying the new bond parameters in program setting panel.

4. Add parameters to the program setting panel

4.1 For HOMA

Insert following code after line 9210:

```

oseLabel = QLabel('O-Se')
oseLabel.setAlignment(Qt.AlignmentFlag.AlignCenter)
homaLayout.addWidget(oseLabel, 13, 0)

oseRopt = QLineEdit()
oseRopt.setText(configFile.get('homa', 'ose').split(',')[0])
homaLayout.addWidget(oseRopt, 13, 2)

oseAlpha = QLineEdit()
oseAlpha.setText(configFile.get('homa', 'ose').split(',')[1])
homaLayout.addWidget(oseAlpha, 13, 4)

```

4.2 For HOMER

Insert following code after line 9287:

```

oseLabelHomer = QLabel('O-Se')
oseLabelHomer.setAlignment(Qt.AlignmentFlag.AlignCenter)
homerLayout.addWidget(oseLabelHomer, 5, 0)

oseRoptHomer = QLineEdit()
oseRoptHomer.setText(configFile.get('homer', 'ose').split(',')[0])
homerLayout.addWidget(oseRoptHomer, 5, 2)

oseAlphaHomer = QLineEdit()
oseAlphaHomer.setText(configFile.get('homer', 'ose').split(',')[1])
homerLayout.addWidget(oseAlphaHomer, 5, 4)

```

5. Set default value

Insert following code after line 9341:

5.1 For HOMA

```

oseRopt.setText('2.0')
oseAlpha.setText('166.0')

```

5.2 For HOMER

```

oseRoptHomer.setText('2.0')
oseAlphaHomer.setText('166.0')

```

6. Save parameters

Insert following code after line 9362:

6.1 For HOMA

```
configFile.set('homa', 'ose', f'{oseRopt.text()}, {oseAlpha.text()}')
```

6.2 For HOMER

```
configFile.set('homer', 'ose', f'{oseRoptHomer.text()}, {oseAlphaHomer.text()}')
```

7. Save and Run

Save the modified `pyaroma_main.py` and run `py.Aroma 5` from source code according to the instruction in Section 1.2.

VI Calculate Dissected NICS with CSGT Method

Using Continuous Set of Gauge Transformations (CSGT) NMR calculations, the individual contributions of each molecular orbital to the NMR shielding tensor could be determined. Similarly, anisotropy of the induced current density (AICD) analysis was employed this method to quantify the contribution of π -orbitals to the ring current. Thus, it is possible to apply CSGT to the calculation of dissected NICS index.

In the case of benzene, MO 17, 20, and 21 were assigned as π -orbitals. An input file for calculating NICS(1) $_{\pi,ZZ}$ of benzene using CSGT method:

```
# NMR=CSGT B3LYP/6-31+G(d) IOp(10/93=2)

Benzene//NICS(1)pi//CSGT

0 1
C      -1.33924  -0.39585  0.00001
C      -0.32669  -1.35773  0.00007
C       1.01242  -0.96188 -0.00006
C       1.3392   0.39597  0.00001
C       0.3268   1.3577   0.00006
C      -1.0125   0.9618  -0.00006
H      -2.38133  -0.70401 -0.00007
H      -0.58109  -2.41425  0.00008
H       1.80038  -1.71023 -0.00015
H       2.38137   0.70386  0.00001
H       0.58095   2.41427  0.00002
H      -1.80028   1.71035 -0.00006
Bq       0.       0.      -1.

benzene_nics_pi_csgt.txt

17
20
21
```

The computed NMR shielding tensors, as reported in the output file, solely originate from contributions of molecular orbitals 17, 20, and 21. These results were compared to those obtained

using the Gauge-Including Atomic Orbital (GIAO) method and the NBO program, as summarized in the following table. Notably, the values obtained from the combined CSGT and GIAO methods, incorporating NBO, fully coincided with those acquired solely from CSGT.

NMR Method	Electrons	NICS(1)	NICS(1) _{zz}
GIAO	σ and π	-10.15	-28.75
CSGT	σ and π	-9.81	-27.09
GIAO ^a	π	-11.84	-29.32
CSGT	π	-10.74	-29.29
GIAO ^a	σ	1.69	0.57
CSGT	σ	0.94	2.20

^aNBO 7.0 was used for CMO analysis.

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User manual of `py.`**Aroma** 5

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